

Review article

Technology review of CNTs TSV in 3D IC and 2.5D packaging: Progress and challenges from an electrical viewpoint

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ABSTRACT

Through-silicon via (TSV) is one of the most important features in 3D integrated circuit (IC) and 2.5D packaging. Both are within the advanced packaging topic for the digital and analog ICs aligned with Moore's paradigm. This article revisits the proposal and progress of carbon nanotubes (CNTs) TSV technology that potentially offers an improvement over the conventional Cu TSV. Today, CNTs TSV has never materialized in commercial products of 3D IC and 2.5D packaging. Compilation on notable numerical modeling works and matching them with related issues in fabrication suggest CNTs TSV technology is still in its infant stage. Although the simulation occasionally shows the advantages of CNTs TSV over Cu TSV in both digital and analog circuits, these results are prone to overestimation. One of the culprits is the number of CNT strands in the bundle which at best can be grown in the fab only $\sim 1\%$ of the theoretically compact bundle used in the *RLC* and *RLGC* models. The direction where CNTs TSV is targeting in 3D IC and 2.5D packaging is not clear by several researchers. As the requirements for high-speed digital and high-frequency analog are different, they are important to be sorted out as an essence of this review to project the path of this CNTs TSV technology.

1. Introduction

1.1. TSV in advanced packaging

The aggressive scaling down of transistors in integrated circuits (ICs), as described by Moore's Law from 1965, refers to the trend of continuously shrinking the size of individual transistors on a microchip [1]. This scaling down allows for more transistors to be packed into the same physical space, leading to increased computing power, improved performance, and energy efficiency. More than Moore's concept which emerged in 2005 acknowledged that not all advancements come from shrinking transistors [2]. Instead, it emphasizes the idea that technological progress is significantly influenced by additional functionalities and innovations beyond traditional scaling. Advanced packaging is one of the tenets of More than Moore aimed at fulfilling the existing and growing demands in consumer mobile electronics, high-performance computing, big data, autonomous vehicles, and the Internet of Things.

Advanced packaging can be divided into two; i) single-die package, and ii) multi-die package. This review specifies our interest in the multi-die package with a core feature of through-silicon via (TSV). Fig. 1 provides the map for this topic, which we can understand the multi-die

package can be further classified as 2D, 2.1D, 2.3D, 2.5D, or 3D depending on their features. Lau explained more on the identification types of advanced packaging in ref. [3]. The vertical connection using TSV reduces the traveling distance for the signal compared to the conventional on-chip interconnect. This gives IC designers more flexibility for heterogeneous integration to create more complex and specialized devices. As a core component in both 2.5D and 3D integration, it is natural for TSV to be one of the most anticipated research topics by people in this field. The TSV technology in this context can be distinguished into two:

- i) TSV at die-level. In the traditional 2D scheme, increasing the transistor density in IC can be achieved by scaling down the transistor footprints. However, there is a point ~ 2 nm where the scaling of CMOS starts to saturate due to material limitations (e.g. quantum tunneling in source, drain, and gate dielectric) and technology limitations (e.g. lithography) [4]. In the state-of-the-art 3D scheme, the number of transistors in a unit area can be increased by stacking the dies and forming TSVs as interconnects. The most successful implementation of 3D IC and its TSV technology can be seen in memory chips. Manufacturers can stack up to 232 layers of non-volatile

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NAND memory cells, and up to 12 layers of dynamic random-access memory cells for the third-generation high-bandwidth memory (HBM3) chip [5]. We observed that Cu TSV is a default technology for the 3D IC.

ii) TSV at package-level. In traditional IC packaging, the footprint of the device is based on the area of the package that encapsulates the chip. An aggressive scaling of the transistor with its related fabrication cost is required for improved performance and power consumption but it does not contribute to the reduction of packaged IC footprint. Dimensions (either 2.5D or 3D) in the context of advanced IC packaging refer to the feature of interposer or substrate where the multichip are bonded. According to Lau in ref. [3], TSV is a feature inside the interposer only for 2.5D and 3D integration, whereas in 2.5D TSV is passive while in 3D TSV is active. For the passive TSV inside the inorganic interposer, TSV with its Si core is fabricated with redistribution layers to complete the chip-chip and chip-substrate interconnects [6]. TSV is considered active when connected to CMOS at the die-level. Hence in the simplest terms, 3D packaging = 3D IC + 2.5D packaging (some also call it 5.5D IC). We observed that Cu TSV is a default technology for the 2.5D packaging.

The ICs can be broadly categorized into two main types based on the type of signals they process; i) digital IC, and ii) analog IC. Additionally, there are mixed-signal ICs that combine both digital and analog components. Both types of IC are relevant to the topic of advanced packaging by:

i) Digital IC. This term refers to ICs that are designed to work with digital signals and perform logical operations. Digital ICs manipulate digital information using binary logic (0 s and 1 s). Some of the well-known examples of digital ICs are microprocessors, memory, application-specific ICs, and field-programmable gate arrays. For examples related to TSV in 3D IC, a 232-layer NAND memory can handle read/write speed 4 GB s^{-1} , while a 12-layer HBM3 chip can handle data speed up to 800 GB s^{-1} per stack. Current technology shows the 3D adaptation with TSV for the digital IC is mainly for memory chips due to the worst heat dissipation and stress for the case of stacked computing ICs [7–9]. These 3D memory ICs with active TSVs can be packaged along with computing ICs on an

interposer as a heterogeneous integration using the 2.5D packaging technology with passive TSVs.

ii) Analog IC. This is a specialized electronic component designed to process continuous electrical signals, typically in the form of varying voltages or currents. Analog ICs are crucial in a wide range of applications where the processing and manipulation of continuous and real-world signals are essential, e.g. audio amplifiers, communication systems, and sensors. RF ICs that operate within the frequency range of 3 kHz to 300 GHz also fall into the parent classification of analog IC. This is a broad range of frequencies that caters to different types of applications, e.g. 2.4 GHz for Bluetooth [10], 24–52 GHz for 5G mobile connectivity [11], 77 GHz for automotive radar [12], 94 GHz for radar imaging [13], and more in $>100 \text{ GHz}$ that are not fully explored. As we learned, the scaling down of analog IC is not as aggressive as digital IC since analog ICs emphasize precision, low noise, and other factors that might not necessarily require the smallest process nodes. Therefore, there is no urgency to market the 3D analog IC using active TSVs. Several kinds of analog IC chips can be placed on an interposer (with passive TSV) and a substrate along with digital IC chips to form a system-in-package [14]. It is a sub-topic of the 2.5D packaging technology towards the miniaturization for a niche application.

1.2. CNTs TSV as a replacement for Cu TSV

At present, Cu is the default material for both active TSV in 3D IC and passive TSV in 2.5D packaging. This makes sense since TSV fundamentally is a part of IC interconnect, whereas interconnect generally refers to the lateral transmission line (TL) made of the Cu layer. The proposal of replacing Cu with specific low-dimensional carbon allotropes, i.e. 1D CNT and 2D graphene can be traced earlier back to 2008 and 2009 when several researchers proposed both of them for IC interconnect [15]. The electron transport in CNT is ballistic as long as the length of CNT does not exceed the mean free path of electrons inside the tube (more in Section 2), thus exerting a minimum resistance as a bundle compared to the conventional metals including Cu. Lower resistance means lower delay and power dissipation in the circuit. A recent compilation of studies in ref. [16] shows CNT measured with ampacity up to $1 \times 10^9 \text{ Acm}^{-2}$ which is much higher than Cu around $8 - 18 \times 10^6 \text{ Acm}^{-2}$.

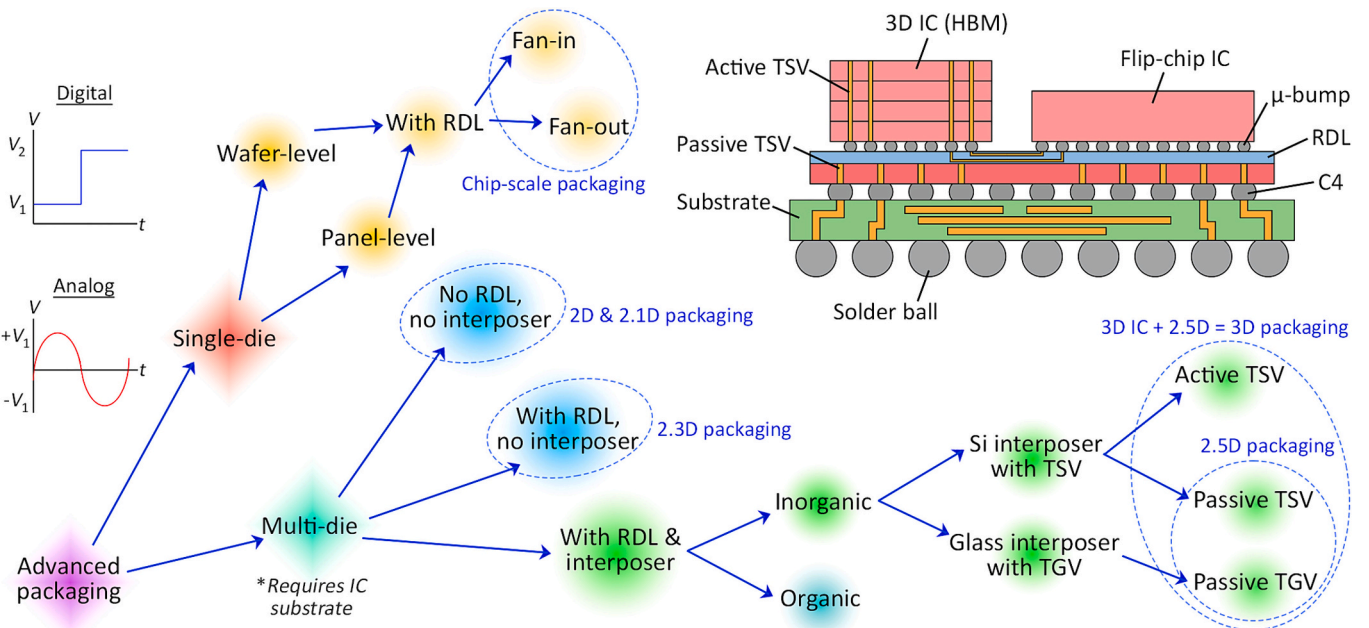


Fig. 1. Position of this TSV topic within the whole body of advanced packaging subject. The device schematic represents the 3D packaging with die-level active TSV and package-level passive TSV. Advanced packaging is applicable for both digital and analog ICs.

Higher ampacity indicates the capability of a material to handle the amount of current before physical failures, e.g. electromigration. The measured thermal conductivity of CNTs is within 2000–3000 $\text{Wm}^{-1}\text{K}^{-1}$ [17], which is significantly higher than Cu $\sim 400 \text{ Wm}^{-1}\text{K}^{-1}$. A high thermal conductivity material is always preferred for building the IC and its package to dissipate the heat, especially for the case of interconnect and TSV where they act as thermal conductor between transistors (hotspots) and large body metal pad (local heat sink). There are several other advantages of CNT in terms of mechanical point of view but not emphasized in this review.

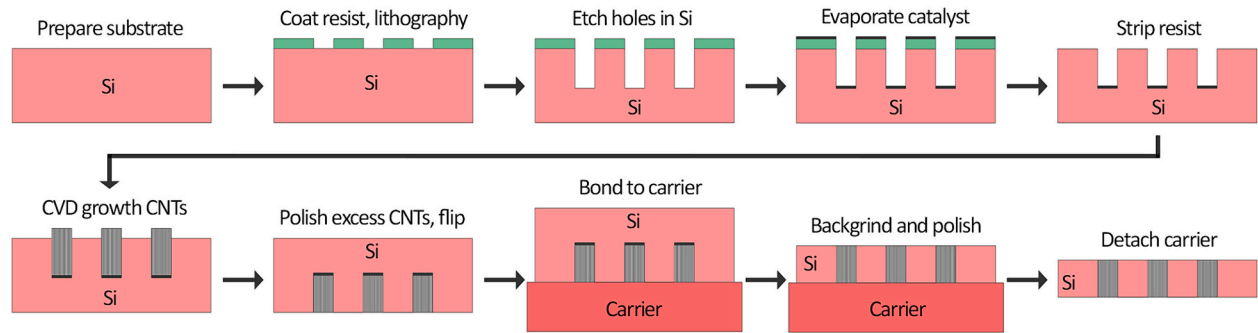
Despite those benefits, the product of using CNT as an interconnect and TSV has never been materialized. CNT has been under the attention of the International Roadmap for Devices and Systems for quite some time, however, this technology is still in the infant stage due to the lack of ground-breaking reports especially answering the problems in processing. On the other hand, Cu is cheap both in terms of the cost of its raw material and processing. We know that Cu is associated with the problem of being dry etched, but the modern semiconductor fab knows very well how to process Cu for IC interconnect and TSV. Therefore for CNTs to even be considered as a replacement of Cu especially for TSV in the context of this review, CNTs TSV simply must promise a better

performance than Cu TSV.

The fabrication of CNTs TSV is quite similar to the Cu TSV, especially in the formation of Si deep holes. The tutorial for forming Cu TSV is available elsewhere [18]. This review will not detail out on the fabrication process, but a quick overview is required to understand the structure and technology of CNTs TSV. We learned there are at least two main approaches for fabricating CNTs TSV; i) direct-growth, and ii) two-wafer. The recipe and mechanism of growing CNTs by chemical vapor deposition (CVD) system is a common review topic and it is available elsewhere [19–21]. The growth of CNTs bundle or forest by low-pressure CVD process essentially requires a catalyst layer on the substrate (commonly Fe), carbon precursor (commonly C_2H_2), and medium-high temperature (around 700–1000 °C), with or without plasma. It is a process for front-end semiconductor manufacturers, but this high-temperature process is only allowed before doping and metallization take place.

Fig. 2a is the direct-growth approach to fabricate CNTs TSV as we redraw according to the process by Wang et al. in ref. [22]. In this process, a deep hole array is patterned by lithography and etched using a deep reactive ion etching (DRIE) system. The metal catalyst which is commonly Fe is evaporated and lifted off to only selectively form the catalyst layer at the bottom of the blind via. Hence after the CVD growth

a First option fabrication of CNTs TSV



b Second option fabrication of CNTs TSV

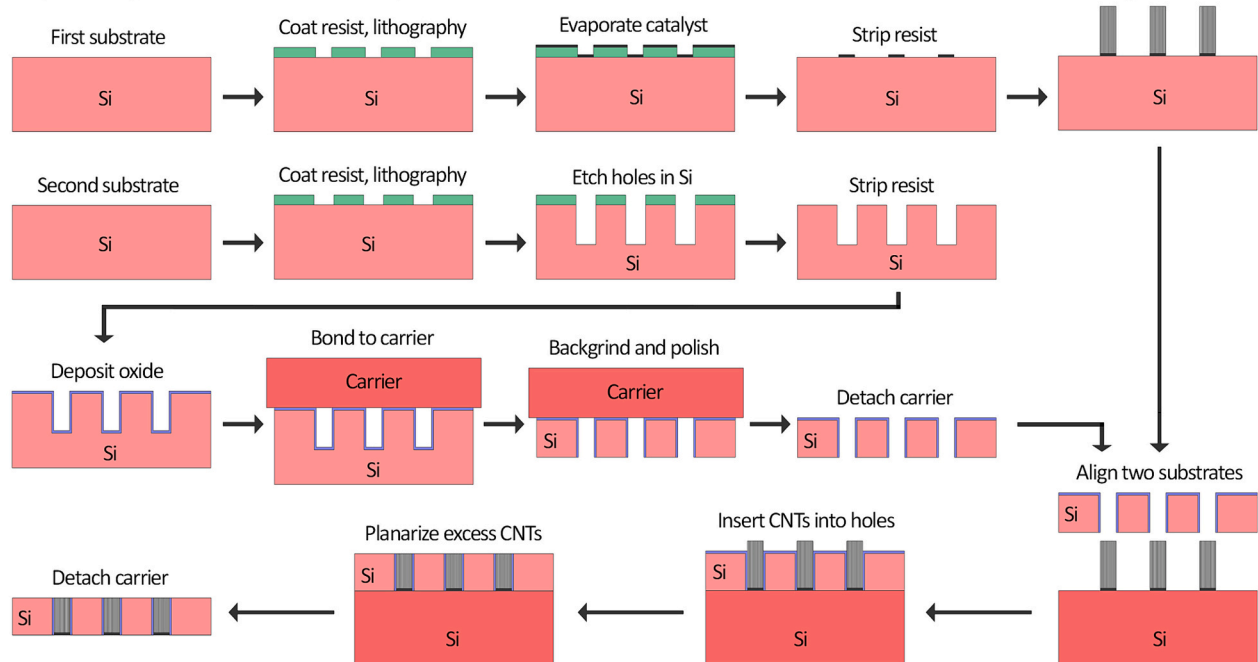


Fig. 2. Popular options on how to fabricate CNTs TSV. **a)** First option by direct growth of CNTs into blind Si via followed by necessary backgrinding and polishing. **b)** Second option by using a two-wafer approach, where CNTs are grown on a first Si substrate, inserted into a pre-fabricated Si through hole, and followed by necessary backgrinding and polishing.

process, the vertically aligned CNTs bundle only grows inside the blind via. This process later undergoes a series of backgrinding and polishing until both sides of the CNTs bundle are exposed for completing the CNTs TSV. Note that the formation of redistribution layers and bumps are not illustrated to only focus on TSV. Apparently, this process is oversimplified and it is likely photoresist needs to be removed first after finishing the DRIE process for adding the plasma CVD process of oxide liner. If we consider the high-temperature CVD process to direct-growth CNTs inside the holes, this approach is then only possible for interposer of 2D packaging with bare Si as a substrate. Exception can be given for 3D IC if this CNTs TSV is fabricated before the chips.

Fig. 2b is the two-wafer approach to fabricate CNTs TSV as we redraw according to the process by Jiang et al. in ref. [23]. In this process, the first substrate (not necessarily the Si wafer) needs to be patterned and grow the CNTs bundle. The second Si substrate is patterned and etched using the DRIE process to form a deep holes array. This substrate then undergoes a series of backgrinding and polishing until both sides of the via are exposed. The most tricky part is to align both wafers to insert the CNTs bundles into through holes. If this process is successful, the excess CNTs will be polished and the first wafer needs to release the CNTs bundle in order to complete the CNTs TSV. Due to the difficulty of inserting CNTs bundle into a through hole, this process is far ready for volume production and limited only to lab-scale investigation. The nature of this technique requires precise alignment and insertion only possible for a sufficiently large structure of 2.5D packaging.

1.3. Objectives and structures of this review

In this article, we conduct a technology review of CNTs TSV by understanding the behavior of CNT and CNTs bundle based on the numerical model, then relate them with concerning issues in fabrication. This kind of systematic review of past publications is always good as it might provide clues on how to solve problems arising in the near future. It is typical for university research to precede industry needs by 10 or more years and potentially there are interesting research papers on CNTs TSV go unnoticed by the community since the industry's attention is concentrated on the 3–5 years span driven by economically motivated goals. Readers may find a short review dedicated to numerical modeling of Cu and CNTs TSVs in ref. [24] useful as a starting point, but viewing this topic purely based on modeling may result in over-expectations of this infant technology. Unfortunately due to the vast area, we cannot fit every aspect of technology in one writing. This review will not cover physical design, failure analysis and reliability, and mechanical aspects of CNTs TSV.

This technology review is conducted systematically to answer three main questions: i) How does CNTs TSV work? ii) Is CNTs TSV a replacement for Cu TSV? iii) What is the next for CNTs TSV? Section 1 provides a compact introduction to the topic for tuning the reader's point of view into macro-to-micro perspective. In Section 2, we compile the progress in understanding the basic properties of CNTs. It is important to start with the basics since CNT is a 1D material that carries distinct electrical properties compared to traditional metals, e.g. Cu. In Section 3, we compile the numerical modeling of CNTs TSV to understand its behavior during its operation carrying the signal. Here, it is critical to identify CNTs either as digital or analog TSVs and relate them to the requirements of each device. We reserved Section 4 for discussing some advanced topics related to CNTs TSV and we conclude this review with our recommendation in Section 5.

2. Electrical properties of CNT for TSV

2.1. Single-strand CNT

A strand of CNT is a 1D material with an exceptionally high aspect ratio. It is not acceptable to treat CNT as a 3D object for fully automated

structural modeling, parameter extraction, and SPICE simulation. The quasi-ballistic carrier transport property of CNT even at room temperature cannot be modeled in the same way we model the metals, e.g. Cu using the known value of bulk resistivity. Fig. 3a shows the magnification view of a CNT under transmission electron microscopy (TEM) with a diameter of around 1–2 nm [25]. The quantum ballistic resistance of a single-strand CNT is known to be:

$$R_Q = \frac{h}{2e^2} \approx 12.9 \text{ k}\Omega \quad (1)$$

where $h = 6.626 \times 10^{-34}$ Js is the Plank constant and $e = 1.602 \times 10^{-19}$ C is the electron charge constant. In the older publications below the year 2010, the resistance of CNT is normally taken at two units of quantum conductance $G_Q = 4e^2/h$, which results in the two-fold lower resistance of $\sim 6.5 \text{ k}\Omega$ [26–28]. For the latest work, we believe R_Q that a single-strand CNT should be taken as one unit G_Q which is similar to the quantization of pristine monolayer graphene [29].

In an experiment, the mean free path of electrons, λ_{mfp} in the 1.8 nm-diameter CNT is measured $\sim 1.6 \mu\text{m}$ [30]. For a single-strand CNT with a length exceeding the value of λ_{mfp} , there is additional ohmic loss due to scattering resistance that needs to be included since the transport shifts from ballistic into a diffusive regime:

$$R_S = R'_S h_{CNT} = \frac{h}{2e^2 \lambda_{mfp}} h_{CNT} \quad (2)$$

where R'_S is the scattering resistance per unit length (p.u.l.) [31], and h_{CNT} is the height of the CNT. An additional metal contact resistance R_{mc} between the metal pad and single-strand CNT needs to be considered for completing a circuit [32]. Therefore, the total resistance for a strand of CNT is:

$$R_{CNT} = R_Q + R_S + R_{mc} \quad (3)$$

In an ab initio simulation, $R_{mc} \approx 3.7 \text{ k}\Omega$ is predicted between Pt and CNT [33]. This value can easily be higher if we consider the poor wettability of metal on CNT and charge depletion inside the CNT itself. For researchers working on numerical modeling of CNT, this contribution R_{mc} is often neglected and some often assign it a specific value depending on their expectations of the fabrication process.

The kinetic inductance L_K in CNT originates from the inertia of the electron mass in motion. When we consider the ballistic transport in CNT, this L_K is notably high because the kinetic energy is not dissipated due to the absence of scattering processes. Assuming $h_{CNT} > \lambda_{mfp}$, there is scattering inside a sufficiently long tube [34]. The inductance of a single-strand CNT is:

$$L_K = L'_K h_{CNT} = \frac{h}{4e^2 \nu_F} h_{CNT} \quad (4)$$

where $L'_K \approx 16.1 \text{ nH } \mu\text{m}^{-1}$ and $\nu_F = 8 \times 10^5 \text{ ms}^{-1}$ is the Fermi velocity of electrons in the CNT [35].

The quantum capacitance C_Q is determined by the extra energy needed to release charge carriers within systems where the energy level of a quantum state is quantized. Again assuming $h_{CNT} > \lambda_{mfp}$, we know C_Q of a single-strand CNT is given by:

$$C_Q = C'_Q h_{CNT} = \frac{2e^2}{h \nu_F} h_{CNT} \quad (5)$$

where $C'_Q \approx 96.8 \text{ aF } \mu\text{m}^{-1}$ is the p.u.l. of C_Q . This value can be slightly lower than the one measured in an experiment [36]. Eqs. 3–5 complete the essential RLC model of the single-strand CNT. Combining them all together, we can write the impedance of the single-strand CNT as:

$$Z_{CNT} = R_Q + (R'_S h_{CNT}) + R_{mc} + j \left(\omega L'_K h_{CNT} - \frac{1}{\omega C'_Q h_{CNT}} \right) \quad (6)$$

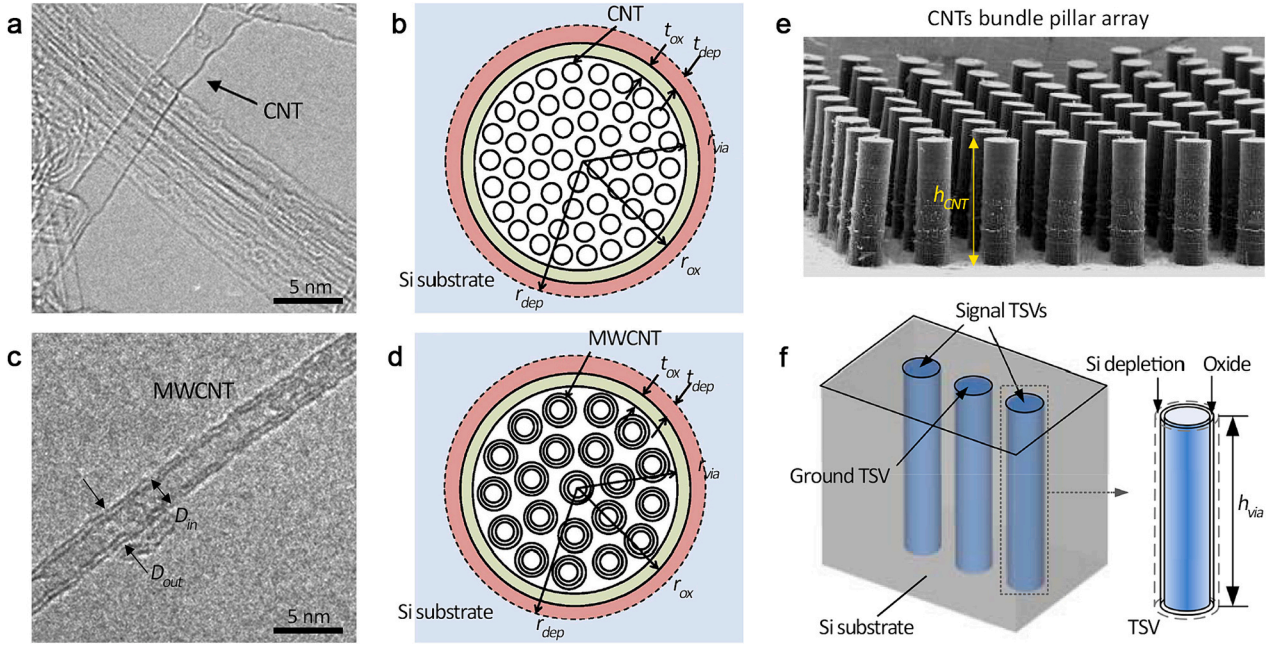


Fig. 3. a,c) Magnification of a CNT and MWCNT (two shells) under TEM. b,d) Schematic of CNTs and MWCNTs bundle filling the cylindrical vias. e) Array of MWCNTs bundle as pillars to indicate the height. f) Schematic of TSV to indicate the height, oxide liner, and Si depletion layer. Images a,c) are from ref. [25], b,d) are from ref. [39], e) is from ref. [58], and f) is from [84].

It is important to distinguish the parameter of p.u.l. for researchers to avoid misinterpretation and error during the numerical modeling. Nevertheless, a single-strand CNT at this point still cannot be used to form TSV due to its 1D geometry. Any possible uses of a single-strand CNT are commonly investigated as interconnect [37] and transistor [38] at a very fundamental level. Therefore, from a commercial perspective, a single-strand CNT currently is not a valuable material for both front-end and back-end semiconductor manufacturers.

2.2. CNTs bundle

Realistically, CNTs are grown in bundles for various types of investigation including for TSV in the context of this review. Fig. 3b shows the schematic of the CNTs bundle inside the cylindrical Si via with oxide liner [39]. By considering the hole is perfectly cylindrical, the maximum number of CNTs in the bundle filling the via can be estimated by:

$$N_{CNT} \approx \frac{2\pi r_{via}^2}{\sqrt{3}(r_{CNT} + \delta_{CNT})^2} \quad (7)$$

where r_{via} is the radius of via, r_{CNT} is the radius of the individual CNT, and $\delta_{CNT} = 0.34$ nm is the minimum Van der Waals intertube spacing for CNT [40]. We need to emphasize that the estimation is based on two assumptions; i) CNT strands are completely packed as a triangle lattice, and ii) the base of substrate for CNTs growing is a perfectly flat circle. Both assumptions, however, are regularly proven to be too optimistic. By taking $r_{CNT} = 1$ nm and $r_{via} = 0.5$ μm , it results in $N_{CNT} \approx 5.05 \times 10^5$ which is equivalent to the theoretical population density of $3.22 \times 10^5 \mu\text{m}^{-2}$. Among the earliest works on CNTs TSV, Kreupl et al. reported the CVD-grown CNTs bundle inside the Si via as shown in Fig. 4a and b. The estimated population density of CNT by the authors is $70 \mu\text{m}^{-2}$, which is equivalent to only 0.02% of the theoretical population density [41]. In more recent work, a high-density CNTs forest grown by Dee et al. with a population density of $4800 \mu\text{m}^{-2}$ can be translated into only 0.95% of the theoretical population density [42].

Note on the second assumption where the base of substrate for CNTs growing is a perfectly flat circle, we keep neutral since it is possible to have this condition. The use of DRIE to form blind holes on Si substrate

(standard technology for Cu TSV) is perfectly fine for an acceptably flat base. However, it is not flat enough to satisfy the assumption to calculate the number of CNT strands in the bundle as described in Eq. 7. Xu et al. employed a two-wafer method to create CNTs TSV where the CNTs bundle can be grown on a polished Si surface [43]. In their report, a polished Si wafer is bonded to the second wafer with pre-fabricated through holes by DRIE. Fig. 4c and d show the CVD growth of the CNTs bundle to complete the CNTs TSV. This method is fine in terms of repeatability and shape of the fabricated CNTs TSV, but the population density of CNTs most likely $< 1\%$ from the theoretical population density as predicted in Eq. 7. Present CVD technology is not capable of growing the theoretical population density as the bundle of CNT strands with $\delta_{CNT} = 0.34$ nm. The nature of CNT is an independent circular wall-type structure, and the surface energy of multiple CNT strands drives them to form into the most stable configuration of separated tubes. Even if the theoretically packed CNTs exist as a bundle, their microscopic appearance will be no longer a tube-like forest but it will be similar to the rigid carbon allotropes such as graphite as the CNT strands are only spaced by the Van der Waals force.

Researchers working on the numerical modeling of the CNTs bundle are aware there is a mix of metallic and semiconducting CNTs within the bundle. The metallic CNT is responsible for the current conduction in the bundle. This is true since CNT can be either metallic or semiconducting depending on its chirality, i.e. armchair is metallic CNT [44], while zigzag is semiconducting CNT [45]. The golden rule of metallic-to-semiconducting ratio $F_m = 1/3$ for the CNTs bundle [46], so far never missed being incorporated into the equation to determine how much CNT is responsible for current conduction. There is an interest from several groups to study the growth of CNTs bundle for preferential growth in armchair configuration [47]. This topic, however, is less active compared to the general topics of growing low-dimensional materials since the controlled growth mechanism for the specific chirality itself is still in debate.

The total number of CNTs in the bundle that are accounted for current conduction is $N_{total} = F_m N_{CNT}$. Consider a case where the CNTs bundle is grown inside the via with $h_{CNT} = h_{via}$, we can write the resistance of the CNTs bundle as:

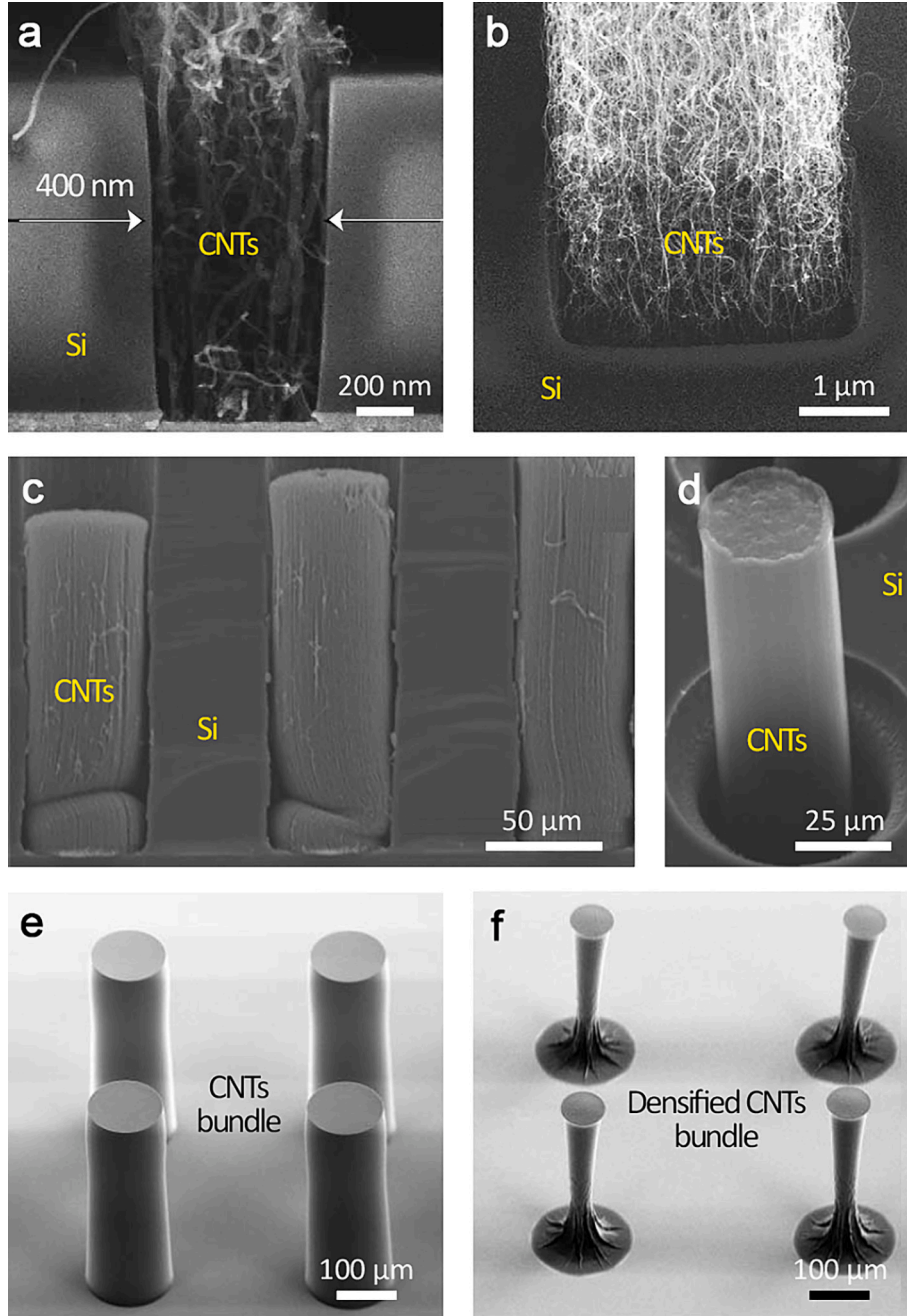


Fig. 4. SEM images of CNTs bundle in Si blind via. **a,b)** Cross-section and tilted view of CNTs bundle with a population density of $70 \mu\text{m}^{-2}$, which we estimate equivalent to only 0.02% of theoretical population density. **c,d)** Cross-section and tilted view of CNTs grown on a polished Si surface using a two-wafer method to create CNTs TSVs. **e,f)** CVD-grown CNTs pillars before and after the densification process. Images **a,b)** are from ref. [41], **c,d)** are from ref. [43], and **e,f)** are from [63].

$$R_{CNTs} = \frac{R_{CNT}}{N_{total}} = \frac{h \cdot h_{via}}{4e^2 \lambda_{mfp} N_{total}} + R_{mc} \quad (8)$$

In Eq. 8, some researchers would like to take R_{mc}/N_{total} after accounting for F_m , but here we can safely put $R_{mc}/N_{total} \approx R_{mc}$ since it depends on the quality of metal contact on the CNTs bundle after the metallization. This R_{mc} can be assigned as a specific value or be ignored during the modeling. Wang et al. fabricated CNTs TSV and they suggest the value of specific contact resistance for Au probe contacting the CNTs bundle TSV is $4.5 \times 10^{-4} \Omega\text{cm}^2$ [48]. Returning to Eq. 8, whenever the conductivity of the CNTs bundle is required to be included in the numerical model, it can be obtained from:

$$\sigma_{CNTs} = \left(\frac{R_{CNTs} \pi r_{via}^2}{h_{via}} \right) \quad (9)$$

The inductance of the CNTs bundle assuming $h_{CNT} = h_{via}$ and $h_{CNT} > \lambda_{mfp}$ can be written as:

$$L_{CNTs} = L'_{CNTs} h_{via} = \left(\frac{L'_K}{2N_{total}} + L'_M \right) h_{via} \quad (10)$$

where L'_M is the mutual inductance p.u.l. arising from the magnetic field by the current flowing through the CNTs in the bundle. It can be written as:

$$L_M = \frac{\mu_0 \ln \frac{h_{via}}{2r_{CNT}}}{2\pi} \quad (11)$$

where μ_0 is the permeability of the vacuum. The magnitude of L_M is related to the aspect ratio of the CNTs bundle. Yu et al. reported L_M is not observed at frequencies up to 10 GHz, and the high-frequency response is effectively damped by the CNTs resistance [49]. In another suggestion, this L_M is relatively small compared to L_K , hence it can be safely discarded during the numerical modeling into $L_{CNTs} = L_K/2N_{total}$ [50]. Note that in the dated paper in ref. [28], they use $L_{CNTs} = L_K/N_{total}$ and this will contribute to a two-fold higher values of L_{CNTs} and L_{CNTs} in the numerical model.

The quantum capacitance p.u.l. of the CNTs bundle for the $h_{CNT} = h_{via}$, as quoted from ref. [51] is:

$$C_{Q-CNTs} = C_Q N_{total} \quad (12)$$

In the CNTs bundle, interactions between individual nanotubes influence the overall capacitance. The proximity of the tubes and the electronic coupling between them contribute to the collective electrical behavior of the bundle. Therefore, the additional p.u.l. electrostatic capacitance of the CNTs bundle can be written as:

$$C_{E-CNTs} = 2C_{En} + \frac{(n_W - 2)}{2}C_{Ef} + \frac{3(n_H - 2)}{5}C_{En} \quad (13)$$

where C_{En} and C_{Ef} are the expressions for the parallel plate capacitances of CNTs with respect to near and far conductors, respectively [31]. In the equation, n_W and n_H represent the number of CNT across the width and length, respectively. They can be approximated as $n_W \approx n_H \approx \sqrt{N_{CNT}}$ for the cylindrical via. The components of C_{En} and C_{Ef} can be detailed as:

$$C_{En} = \frac{2\pi\epsilon_0}{\ln(d_{via}/d_{CNT})} \quad (14)$$

$$C_{Ef} = \frac{2\pi\epsilon_0}{\ln(2d_{via}/d_{CNT})} \quad (15)$$

where ϵ_0 is the vacuum permittivity, d_{via} is the diameter of the via, and d_{CNT} is the diameter of the individual CNT residing in the via. Finally, we can write the capacitance of the CNTs bundle as:

$$C_{CNTs} = C_{CNTs} h_{via} = \left(\frac{1}{C_{Q-CNTs}} + \frac{1}{C_{E-CNTs}} \right)^{-1} h_{via} \quad (16)$$

Considering the C_Q per channel is $2 \times (2e^2/h\nu_F) \approx 193$ aF μm^{-1} , researchers working on modeling of CNTs bundle often ignore C_{Q-CNTs} in the formula [52–54]. In fact, if we compare it to the oxide capacitance and depletion layer in Si, later we can also neglect the contribution of C_{E-CNTs} in the formula [55]. Combining Eqs. 8, 10, and 16 (still retaining the contribution of capacitance), we can write the impedance of the CNTs bundle as:

$$Z_{CNTs} = R_{CNTs} + j \left(\omega L_{CNTs} - \frac{1}{\omega C_{CNTs}} \right) \quad (17)$$

At this point, the *RLC* model for the CNTs TSV is completed but Z_{CNTs} in Eq. 17 needs to be discussed more to clear up some confusion. They are summarized as follows:

- i) Considering we have a CNTs TSV with a structure of CNTs (signal)-oxide(liner)-Si(substrate), we agree that R_{CNTs} and L_{CNTs} are the only parameter that researcher should include in their model. Since the contribution of capacitance and from oxide and Si depletion layer is more important, the capacitance of TSV can be written as $C_{TSV} = (C_{ox}^{-1} + C_{dep}^{-1})^{-1}$ without C_{CNTs} . This will be

relevant in Section 3 for the high-frequency application such as 2.5D packaging of RF components.

- ii) Regardless, the impedance of the CNTs bundle still follows the one in Eq. 17. This *RLC* model for the CNTs bundle is helpful for simple a circuit simulation for a single CNTs TSV. Subash et al. employed this *RLC* model for the CNTs bundle as an interconnect, where they focused the simulation on varying the length of CNTs and F_m to understand their effect on R , L , and C parameters [56]. In this case, C_{TSV} is not important since it is not a complete circuit simulation.
- iii) Considering this CNTs TSV is for the digital signal for speed up to a few GHz, L_{CNTs} is then not important. Here, R_{CNTs} , C_{CNTs} , and C_{TSV} are important for a complete *RC* model. Researchers would find this model is sufficient to be used in a driver-TSV-load setup for studying the delay and power dissipation from CNTs TSV during the transmission of digital signals. This is most relatable for 3D digital IC.

2.3. MWCNTs bundle

Conceivably as a more promising material than CNT due to its higher current carrying capability, the MWCNT is a thicker version of CNT which consists of multiple concentric layers. Fig. 3c shows the example of a magnification view taken on a two-shell MWCNT under TEM [25]. The number of shells (walls) in the MWCNT can be determined by:

$$N_{shell} = 1 + \frac{D_{out} - D_{in}}{2\delta_{CNT}} \quad (18)$$

where D_{out} is the outer-most diameter, D_{in} is the inner-most diameter of a single-strand MWCNT, and $\delta_{CNT} = 0.34$ nm is a Van der Waals inter-tube spacing. Fig. 3d shows the schematic of the MWCNTs bundle inside the cylindrical Si via with oxide liner [39].

The resistance of MWCNTs is quite complicated to be modeled compared to the CNTs bundle. The number of conducting channels and λ_{mfp} depends on the number of shells, the diameter of each shell, and the temperature. In contrast to the CNT, the zigzag configuration (semi-conducting) can contribute to the conduction due to the large diameter of the MWCNT. The number of conducting channels can be written as:

$$M_{shell} \approx \begin{cases} 2/3, & D < d_T/T \\ \approx aD.T + b, & D > d_T/T \end{cases} \quad (19)$$

where $a = 2.04 \times 10^{-4}$ nm⁻¹K⁻¹, $b = 0.425$, $d_T = 1300$ nm.K, and T is the temperature [57]. The value of M_{shell} from Eq. 19 will represent the average number of conducting channels for each shell (it can be either metallic or semiconducting chirality). This value stands on the assumption that each shell has a probability of being metallic chirality is 1/3, which is similar to the statistic value of F_m for the CNTs bundle as we discussed in the previous section. This condition is known as valid for MWCNT under small voltage bias typically for interconnect analysis.

Consider a strand of MWCNT with a length of $h_{MWCNT} = h_{via}$. Hence, we can write the resistance of a single-strand MWCNT as:

$$R_{MWCNT} = R_Q + R'_S h_{via} + R_{mc} = \left(\frac{h}{2e^2} + \frac{h \cdot h_{via}}{2e^2 \lambda_{mfp}} \right) \frac{1}{M_{shell}} + R_{mc} \quad (20)$$

To achieve a specific value of $h_{CNT} = h_{via}$ for complete filling the via with MWCNTs bundle, researchers may utilize the multiple stacked growths on the MWCNTs bundle. Li et al. demonstrated the feasibility of this growth technique to stack up to eight layers of MWCNTs bundle, forming the pillar array as shown in Fig. 3e with h_{CNT} up to 1.8 mm [58]. Depending on the fabrication process, either by using direct growth inside the blind via or by precision slot-in two-wafer method, we foresee the condition of $h_{CNT} = h_{via}$ for obtaining MWCNTs TSV as illustrated in Fig. 3f is feasible to be achieved.

In Eq. 20, since the diameter of CNT is increased, the estimation of $\lambda_{mfp} \approx 1.6$ μm for the 1.8 nm-diameter CNT used in Eq. 2 is no longer

valid for MWCNT. The linear relationship with diameter is known to be $\lambda_{mfp} = 10^3 D(T_1/T)$, where $T_1 = 400$ K is the reference value for temperature [57]. Fig. 3d shows the schematic of the MWCNTs bundle inside the cylindrical Si via with oxide liner [39]. For the case of MWCNTs bundle filling the via, we first need to determine the N_{MWCNT} using the geometrical relation for the cylindrical via similar to the one described in Eq. 7 for the CNTs bundle. The resistance of the MWCNTs bundle inside a via then can be written as:

$$R_{MWCNTs} = \frac{R_{MWCNT}}{M_{shell}} = \left(\frac{h}{2e^2} \frac{h \cdot h_{via}}{2e^2 \lambda_{mfp}} \right) \cdot \frac{1}{N_{MWCNT} M_{shell}} + R_{mc} \quad (21)$$

In both Eqs. 20 and 21, we isolate R_{mc} from taking account of M_{shell} and N_{shell} as it is heavily depended on the quality of metallization. Ghosh et al. found $R_{mc} = 35.15$ k Ω that the Cu contacted the MWCNTs bundle [59], while Fiendler et al. observed $R_{mc} = 11 \pm 9$ k Ω that the Ta contacted the MWCNTs bundle [60]. This kind of information from experiments is valuable to be included during the numerical modeling of MWCNTs TSV. Up to this point, we can take the value of R_{MWCNTs} to obtain the σ_{MWCNTs} using Eq. 9 by accounting for the geometry of the cylindrical via.

Xie et al. fabricated the MWCNTs ($N_{shell} = 3 - 8$, $D_{out} = 5 - 13$ nm) TSV with a geometry of $r_{via} = 2.5$ μ m and $h_{via} = 25$ μ m [61]. They observed an ohmic behavior for the MWCNTs probed using tungsten tips and they measured $R_{MWCNTs} = 297$ Ω which is equivalent to $\sigma_{MWCNTs} = 43$ Scm $^{-1}$. Again, these values are considerably high for MWCNTs to be successfully considered as a conductor in TSV. The culprit is the packing density of MWCNTs inside the via as defined in Eq. 7 is hardly relevant to the actual grown MWCNTs bundle, hence affecting the N_{MWCNT} and R_{MWCNTs} . There is a technique known as the capillary densification process introduced by De Volder et al. to pack up the bundle of MWCNTs [62]. The essence of this technique is to expose the CNTs or MWCNTs bundle to the boiling acetone and utilize the capillary force from the condensation to form a high-density pillar.

Wang et al. fabricated MWCNTs TSV with a geometry of $r_{via} = 50$ μ m and $h_{via} = 195$ μ m [63]. After a densification process using acetone vapor, they obtained $R_{MWCNTs} = 2.7 \pm 0.5$ Ω which is equivalent to $\sigma_{MWCNTs} = 256$ Scm $^{-1}$. This is good progress for researchers working on CNTs or MWCNTs TSV, but we have to understand the densification process did not increase the total number of strands within the bundle. In Fig. 4e and f, the densified bundle will appear with a smaller diameter, hence there will be a void inside the TSV. Li et al. use the expression of R_{MWCNTs} in Eq. 21 to build a driver-via-load setup for circuit simulation in IC interconnect with vias [64]. They showed how the delay is proportional to R_{MWCNTs} compared to the known technology of Cu, where they anticipated a serious delay when a low-density MWCNTs bundle is used to replace Cu in global via (top-most level vias). This kind of feedback from simulation needs to be aware by the process development team for the successful proposal of this CNTs TSV technology.

Almost all discussion on fabricated MWCNTs TSV concludes at R_{MWCNTs} . The remaining components of inductance and capacitance will need to be learned from the numerical modeling. The inductance in the MWCNTs bundle is a sum of kinetic inductance and mutual magnetic inductance between different shells. As quoted from ref. [65], it can be written as:

$$L_{MWCNTs} = \left(\frac{L_K}{N_{MWCNT}} + L_{M_MWCNTs} \right) h_{via} \quad (22)$$

where the mutual magnetic inductance p.u.l. of the MWCNTs bundle is given by:

$$L_{M_MWCNTs} = \frac{\mu_0}{2\pi} \left(\ln \frac{4h_{CNT}}{D_{out}} - 1 + \frac{D_{out} + D_{in}}{\pi h_{CNT}} \right) \quad (23)$$

The capacitance in the MWCNTs bundle is a sum of quantum capacitance and electrostatic capacitance in series as follows:

$$C_{MWCNTs} = C_{MWCNTs} h_{via} = \left(\frac{1}{C_{Q_MWCNTs}} + \frac{1}{C_{E_MWCNTs}} \right)^{-1} h_{via} \quad (24)$$

We can ignore C_{Q_MWCNTs} for the quantum capacitance p.u.l. as the channel number in the MWCNTs bundle is very large enough compared to its electrostatic element [53]. The electrostatic capacitance p.u.l. of the MWCNTs bundle, as quoted from ref. [65] can be written as:

$$C_{E_MWCNTs} = \frac{2\pi\epsilon_0}{\ln(D_{out}/D_{in})} = \frac{2\pi\epsilon_0}{\ln[D_{out}/(D_{out} - 2\delta_{CNT})]} \quad (25)$$

Hence, the impedance of MWCNTs bundle TSV can be written as:

$$Z_{MWCNTs} = R_{MWCNTs} + j \left(\omega L_{MWCNTs} - \frac{1}{\omega C_{MWCNTs}} \right) \quad (26)$$

In another interest of utilizing MWCNT as a lateral interconnect, this *RLC* model is good enough to be transformed into an equivalent single conductor (ESC) model [66,67], or multi-ESC model [68,69] to reduce computational resources. While the ultimate goal of the TL model and ESC model is the same for the circuit simulation of MWCNT, the origin of the ESC model is actually to simplify the *RLC* based on the number of shells inside the single-strand MWCNT by assuming; i) all shells are parallel and connected at both ends, and ii) when the MWCNT interconnect is stimulated in a common mode configuration, the voltages at arbitrary cross-sections exhibit uniform values. As the ESC model for the single-strand MWCNT can be extended for the MWCNTs bundle, the difference between TL and ESC models becomes unnoticeable, and sometimes quoted as the same thing [70].

The impedance of CNT, either in the forms of single-strand, CNTs bundle, or MWCNTs bundle, has been modeled by many researchers using the methodology described in this section. Due to its unique 1D structure, CNT needs to be modeled firstly by understanding its ballistic-diffusive carrier transport, then relate them to the experimental reports. We found a shortcoming in modeling is due to the calculation of the number of strands in a bundle which is mentioned in Eq. 7. Relying only on the theoretical values of N_{CNT} and N_{MWCNT} are the origin of errors in modeling of Z_{CNTs} and Z_{MWCNTs} . Compact packing of strands in a bundle according to the theoretical limit (spacing only $\delta_{CNT} = 0.34$ nm) so far has never been achieved by material scientists. If there is some correction factor for both N_{CNT} and N_{MWCNT} can be introduced, then the built *RLC* model is good enough to study the behavior of a single CNTs TSV either for digital or analog circuits. For example in Fig. 5a and b, researchers working on digital signals would find this model is sufficient to be used in a driver-TSV-load setup for studying the delay and power dissipation from a single CNTs TSV in 3D IC. The correction factor for both N_{CNT} and N_{MWCNT} can be included in the numerical model as a multiplier to match with the capability of the CVD method to grow CNTs bundle, e.g. $\gamma_{CVD} = 0.01$ multiplier for 1% of the theoretically N_{CNT} and N_{MWCNT} .

3. Behavior of multiple CNTs TSVs

3.1. Simple RLGC model for a pair of TSV

The first three components of *RLC* are described in the previous Section 2 attributed to the material and geometrical properties of CNTs filling the cylindrical hole. The remaining components of *GC* need to be modeled for a complete condition of Si via with oxide liner. The capacitance that arises between the two CNTs TSVs is due to the conductive Si substrate. It can be determined using the parallel-wire capacitance model [71], which yields to:

$$C_{Si} = \frac{2\pi\epsilon_{Si}h_{via}}{\cosh^{-1} \left(\frac{p_{via}^2}{2(r_{via} + t_{ox} + t_{dep})} - 1 \right)} = \frac{\pi\epsilon_{Si}h_{via}}{\cosh^{-1} (p_{via}/(2r_{dep}))} \quad (27)$$

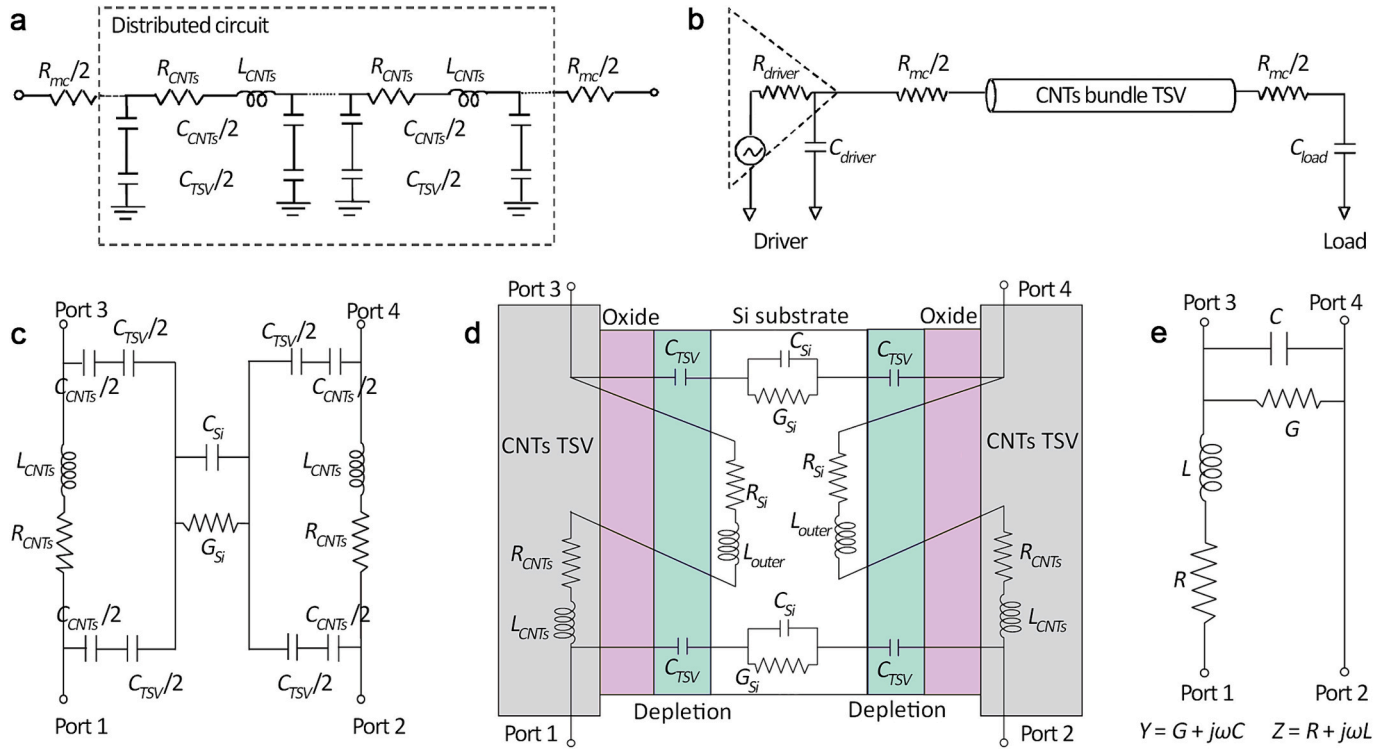


Fig. 5. a) RLC model and its equivalent circuit for the CNTs bundle TSV, and b) implementation of the RLC model into driver-TSV-load setup. c) RLGC model for a pair of CNTs TSV with extra consideration of eddy current inside the Si substrate, d) simplified schematic for the pair, and e) simplest RLGC representation for the pair. Schematics in a,b) are taken from ref. [34], and c,e) are taken from ref. [71].

where ϵ_{Si} is the permittivity of the Si, p_{via} is the center-to-center distance between adjacent TSVs, r_{via} is the radius of via, t_{ox} is the thickness of oxide, t_{dep} is the thickness of the depletion layer, and r_{dep} is the radius of the depletion layer in Si. As suggested in ref. [72], it is recommended to simplify the t_{dep} by considering the maximum depletion width of a planar MOS capacitor by:

$$t_{dep} = \sqrt{\frac{4\epsilon_{Si}k_B T}{e^2 N_a} \ln\left(\frac{N_a}{n_i}\right)} \quad (28)$$

where $k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$ is the Boltzmann constant, N_a is the acceptor concentration in a p-type Si, and $n_i = 1.08 \times 10^{10} \text{ cm}^{-3}$ is the intrinsic carrier concentration for Si.

The conductance of the Si substrate can be obtained by relating to C_{Si} :

$$G_{Si} = \sigma_{Si} C_{Si} / \epsilon_{Si} \quad (29)$$

where σ_{Si} is the bulk conductivity which can be further detailed as a function of doping and temperature. Take the specific value of N_a for the Si substrate, we can write $\sigma_{Si}(T)$ as:

$$\sigma_{Si} = eN_a \left[\mu_{min} + \frac{\mu_{max} - \mu_{min}}{1 + (N_a/N_{ref})^{0.76}} \right] \cdot \left(\frac{T}{300 \text{ K}} \right)^{-1.5} \quad (30)$$

where the min and max holes mobility of Si are taken as $\mu_{min} = 47.7 \text{ cm}^2 \text{ V}^{-1} \text{ S}^{-1}$ and $\mu_{max} = 495 \text{ cm}^2 \text{ V}^{-1} \text{ S}^{-1}$, and reference doping $N_{ref} = 6.3 \times 10^{16} \text{ cm}^{-3}$ [73]. In a condition where the high-frequency RF passes through the TSV, it will generate a varying magnetic field. An eddy current is expected to be circulating inside the Si substrate with $\sigma_{Si_eff} = \sigma_{Si} + j\omega\mu\epsilon_{Si}$, where ω is the angular frequency [74]. There are rising research works on investigating the use of glass ($\sigma_{glass} = 1 \times 10^{-14} \text{ Sm}^{-1}$) substrate replacing Si substrate and oxide liner to form through glass via (TGV) [75–77]. Glass substrate for TGV does not

require an oxide liner and there is no depletion layer inside the glass due to the MOS effect. Since glass is a very poor electrical conductor, the parasitic GC between a pair of TGV by C_{glass} and G_{glass} are suppressed compared to the conventional Cu TSV and CNTs TSV. The review of the advantages and disadvantages of TGV compared to TSV, however, is beyond the scope of this article.

Note that in the circuit of two adjacent TSVs, we have the parallel connection of C_{Si} and G_{Si} where they are then connected to the individual capacitance of TSV as illustrated in Fig. 5c. The capacitance p.u.l. for the TSV with oxide liner can be derived by applying the coaxial-wire capacitance model:

$$C'_{TSV} = \left(\frac{1}{C_{ox}} + \frac{1}{C_{dep}} + \frac{1}{C_{CNTs}} \right)^{-1} \approx \left(\frac{1}{C_{ox}} + \frac{1}{C_{dep}} \right)^{-1} \quad (31)$$

where C_{ox} is the capacitance of the oxide liner and C_{dep} is the capacitance of the Si depletion region. Despite being elaborated above in Eqs. 16 and 24, the contribution of capacitance in CNTs and MWCNTs bundle can be safely discarded because its value is sufficiently large compared to the oxide capacitance and Si depletion layer. It also means the expression for C'_{TSV} is not unique for CNTs TSV but similar to any materials, e.g. Cu TSV. The components of C_{ox} and C_{dep} can be detailed as:

$$C_{ox} = \frac{2\pi\epsilon_{ox}h_{via}}{\ln(r_{ox}/r_{via})} \quad (32)$$

$$C_{dep} = \frac{2\pi\epsilon_{Si}h_{via}}{\ln(r_{dep}/r_{ox})} \quad (33)$$

where ϵ_{ox} is the permittivity of the oxide, r_{ox} is the radius of the oxide, and ϵ_{Si} is the permittivity of the Si. The origin of C_{dep} is from the MOS effect, although the Si substrate is either intentionally or unintentionally biased during the operation. Hence similar to MOSFET analysis, the width of the depletion region can be further detailed in the consideration of weak and strong inversions due to the potential in the TSV [78].

Finally, we can write C_{TSV} from Eq. 31 into:

$$C_{TSV} = \left[\frac{1}{2\pi\epsilon_{ox}} \ln \left(\frac{r_{via} + t_{ox}}{r_{via}} \right) + \frac{1}{2\pi\epsilon_{Si}} \ln \left(\frac{r_{via} + t_{ox} + t_{dep}}{r_{via} + t_{ox}} \right) \right]^{-1} h_{via} \quad (34)$$

Combining the RLC model for the individual CNTs TSV in Section 2 with the GC model above completes the simple $RLGC$ model for a pair of CNTs TSVs. This model can be used to analyze both digital and analog signals passing through the CNTs TSVs. There is no need to discuss in detail the simulation data presented by every researcher working on $RLGC$ modeling of multiple CNTs TSVs as the CNT here is not the one that plays the deciding role. The design aspects especially the number of TSV, pitch, and substrate material are the main considerations determining the performance of multiple CNTs TSVs. If there are any changes in R , L , and C of the CNTs bundle, the behavior of a pair of CNTs TSVs (after being $RLGC$ modeled and simulated) indeed will show changes but again that is something we already can expect. For example in a report by Zheng et al. [55] where they conducted this kind of simple $RLGC$ modeling and simulation for RF signal up to 100 GHz, the conclusion of F_m affecting the transmission is already according to everyone's expectation since F_m directly influences the value of R . If F_m of CNTs is too low (less amount of metallic CNTs in the bundle), the transmission can be worse than Cu so there is no point to replace Cu TSV with CNTs TSV.

3.2. Improved $RLGC$ model for coupled CNTs TSVs

The simple $RLGC$ model in Section 3.1 can be used to analyze both digital and analog signals passing through a pair or multiple CNTs TSVs. However, it is not accurate especially for a circuit simulator of high-frequency RF signals since did not consider the skin effect in CNTs TSV and eddy current in Si substrate. For this purpose, we refer to the $RLGC$ model introduced by Xu et al. in ref. [79] for the improved circuit model. Fig. 5d shows the schematic of two adjacent CNTs TSVs and the components of R , L , G , and C by including the contribution of skin effect and eddy current. We can express the inner via impedance p.u.l. of the CNTs or MWCNTs bundle inside the via as:

$$Z_{via} = \frac{(1-j) \cdot J_0((1-j)r_{via}/\delta_{TSV})}{\sigma_{CNTs} \cdot 2\pi r_{via} \delta_{TSV} \cdot J_1((1-j)r_{via}/\delta_{TSV})} \quad (35)$$

where J is the Bessel function of the first type ($J_0 = 0^{\text{th}}$ order, $J_1 = 1^{\text{st}}$ order). The skin depth in CNTs TSV is given by $\delta_{TSV} = \sqrt{2/\omega\mu_0\sigma_{CNTs}}$. Note that $\omega = 2\pi f$ is the angular frequency of the signal. Next, the resistance p.u.l of Si and outer inductance p.u.l. of the CNTs TSV, as quoted from ref. [71] can be written as:

$$R_{Si} \approx \frac{\omega\mu_0}{2} \cdot Re \left[H_0^{(2)} \left(\frac{(1-j)r_{dep}}{\delta_{Si}} \right) \right] - \frac{\omega\mu_0}{2} \cdot Re \left[H_0^{(2)} \left(\frac{(1-j)p_{via}}{\delta_{Si}} \right) \right] \quad (36)$$

$$L_{outer} \approx \frac{\mu_0}{2} \ln \left(\frac{r_{dep}}{r_{via}} \right) + \frac{\mu_0}{2} \cdot Im \left[H_0^{(2)} \left(\frac{(1-j)r_{dep}}{\delta_{Si}} \right) \right] - \frac{\mu_0}{2} \cdot Im \left[H_0^{(2)} \left(\frac{(1-j)p_{via}}{\delta_{Si}} \right) \right] \quad (37)$$

where $H_0^{(2)} = 0^{\text{th}}$ is the Handel function of the second type. The skin depth in Si is given by $\delta_{Si} = \sqrt{2/\omega\mu_0(\sigma_{Si} + j\omega\epsilon_{Si})}$. Fig. 5e is the simplest form of the $RLGC$ model to represent Eqs. 35–37. The effective serial impedance $Z = R + j\omega L$ and effective parallel admittance $Y = G + j\omega C$ of a pair of CNTs TSV can be written as:

$$Z_{eff} = R_{TSV} + j\omega L_{TSV} = (2Z_{via} + R_{Si} + j\omega L_{outer})h_{via} \quad (38)$$

$$Y_{eff} = \left(\frac{1}{2(j\omega C_{CNTs})} + \frac{1}{2(j\omega C_{TSV})} + \frac{1}{(j\omega C_{Si} + G_{Si})} \right)^{-1} \quad (39)$$

Note that the C_{CNTs} , C_{TSV} , C_{Si} , and G_{Si} are already included the contribution of h_{via} , so that both Z_{eff} and Y_{eff} in Eqs. 38 and 39 are not in the form of p.u.l. Using the improved $RLGC$ model, researchers can investigate the behavior of two or more CNTs TSVs.

In this review, we see the importance of this improved $RLGC$ modeling to understand the CNTs TSV technology, but still, there is no need to discuss in detail the simulation data presented by every researcher as CNT is not the one that plays the deciding role. The design aspects especially the number of TSV, pitch, and substrate material are the main considerations determining the performance of multiple CNTs TSVs. Digital or analog signals within the electronics system can be transmitted either by common-mode (CM) or differential-mode (DM) signaling approaches. The use of the DM technique is well known to improve signal integrity, reduce electromagnetic interference, and enhance noise immunity in high-speed digital data transmission and high-frequency analog interconnect. A basic unit of DM CNTs TSV is a pair of CNTs TSV that carry the complementary signals S^+ and S^- . The configurations of CNTs TSV after completing the $RLGC$ modeling can be summarized as follows:

- i) Three linear arrangement TSVs, which can be either SSS-type, SGS-type, or GSG-type of TSVs. This is an extension for the two via problem in above and it represents the repeating array of TSVs. Commonly two aggressors and one victim are studied for the SSS-type of CNTs TSV [74,80], and one aggressor and one victim for the SGS-type of CNTs TSV.

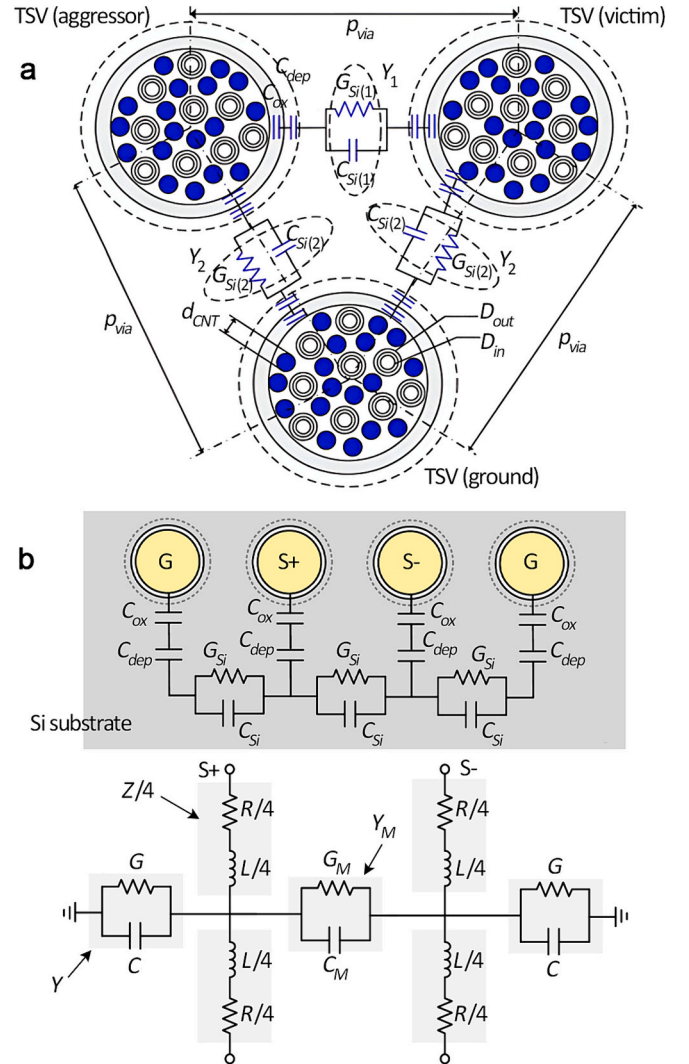


Fig. 6. a) Three triangular arrangement SSG-type CNTs TSVs, where two signal TSVs can be either in CM signals S^+S^+ or DM signals S^+S^- . b) Four linear arrangement GSSG-type CNTs TSVs, where it can be CM signals GS^+S^+G or DM signals GS^+S^-G . Images a) is from ref. [81], b) is from ref. [83].

- ii) Three triangular arrangement SSG-type TSVs as shown in Fig. 6a, where two signal TSVs can be either in CM signals S^+S^+ or DM signals S^+S^- . A single ground TSV is shared and located near these two signal TSVs to form a triangle arrangement. The common mode S^+S^+ is a default problem to be considered [71], but is known to be more susceptible to noise coupling compared to the DM signals S^+S^- . Generally, DM signaling can be considered an advanced topic for TSV, but it is not unique to the utilization of CNTs as the conductor. Numerical modeling and analysis for the DM CNTs TSV for this configuration are available in refs. [81, 82].
- iii) Four linear arrangement GSSG-type TSVs as shown in Fig. 6b, can be CM signals GS^+S^+G or DM signals GS^+S^-G . Numerical modeling for the metals and CNTs TSV for this configuration is available in ref. [83].
- iv) Five triangular arrangement SSSSG-type TSVs, where two of signal TSVs are a pair and it can be either in CM signals S^+S^+ or DM signals S^+S^- . A single ground TSV is shared and located in the middle of these two pairs of signal TSVs. Numerical modeling for this configuration is available in ref. [84].

The digital interconnect may use high-speed current steering logic (HCSL) as the CM approach. HCSL typically uses a single-ended signaling scheme where the data is encoded in the voltage levels of a single signal line. The voltage swing of HCSL signals is often around a specified common-mode voltage level (e.g. around 0.8–1.2 V). Single-ended signaling can simplify the design of certain circuits but may be more susceptible to noise compared to differential signaling. The use of the DM approaches by high-speed digital interconnect to use positive emitter-coupled logic (PECL) and current mode logic (CML) are more common compared to HCSL. Both PECL and CML are known for their high-speed operation and are used in applications where high performance and speed are critical. Unfortunately, the numerical modeling and simulation works (either CM or DM) on CNTs TSV rarely focus on the digital circuit so the important indicators, e.g. jitter, duty cycle distortion, crossing, and eye diagram are not well understood to be compared with Cu TSV. Many reports prefer to focus on high-frequency analog where S parameters from multiple CNTs TSVs are regularly discussed and compared with Cu TSVs.

We notice that the progress on numerical modeling of CNTs TSV is most active within 2010–2016 and started to slow down in recent years until 2024. This slowing down in progress means the fundamentals for building the RLGC are already well established for CNTs TSV, so the bottleneck for progressing this technology is at the fabrication part. Given the properties of the CNTs bundle after the fabrication are significantly different from the model (especially on N_{CNT} that governs R , L , and C of the CNTs bundle), then one would switch to a doubtful point of view which puts the numerical modeling of CNTs TSV into questionably accurate in predicting the behavior of actual fabricated CNTs TSV. Most of the validation by researchers means they observe a good fitting between the manual circuit calculations with the commercial circuit simulation tools, e.g. Ansys HFSS [85], CST Studio Suite [86], and COMSOL Multiphysics [87]. They support automation for the conventional Cu TSV starting from 3D modeling, parameter extraction, and circuit simulation for accelerating the R&D works. These features, however, are not applicable for CNTs TSV (refer to workflow in Fig. 7) unless the researcher resorts to the assumption that the CNTs bundle is a solid conductor similar to Cu. Validation of manual circuit simulation with commercial circuit simulation for CNTs TSV is a promising research direction. However, any discrepancies between manual and automated solutions do not necessarily indicate a highly accurate representation of actual fabricated CNTs TSV. This is because both solutions utilize the same RLGC model, which has been developed for the 1D CNT and its bundle.

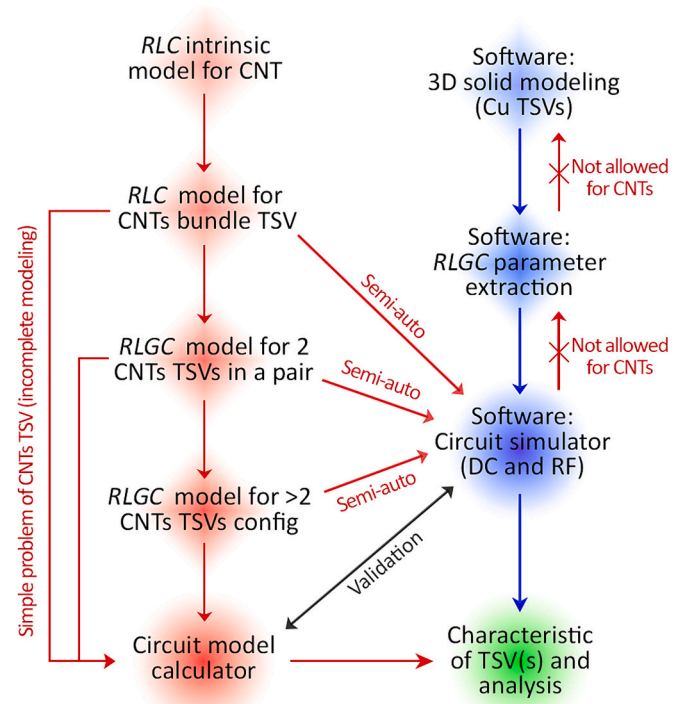


Fig. 7. Workflow of the numerical modeling of CNTs TSV. The figure indicates the options for a semi-automated approach and validation using the commercial circuit simulation tool.

4. Advanced topics of CNTs TSV

4.1. Tapered via

In the previous Sections 2 and 3, the numerical modeling works are based on the condition of CNTs or MWCNTs residing in a perfect cylindrical via. However, during the fabrication of via especially for the relatively large via interposer by DRIE process, the formation of the hole is natural to be positively tapered with the angle below 90° . Assuming the case of CNTs bundle is closely packed in the tapered via, the number of CNTs in the bundle filling the tapered cylindrical via, as quoted from ref. [88] can be written as:

$$N_{CNT_tapered} = \frac{\frac{\pi}{3} (r_{a_via}^2 + r_{b_via}^2 + r_{a_via} \times r_{b_via})}{\frac{2}{\sqrt{3}} (r_{CNT} + \delta_{CNT})^2} \quad (40)$$

where r_{a_via} is the lower radius (small) and r_{b_via} is the upper radius (large) for the positively tapered cylindrical via. After the value of $N_{CNT_tapered}$ is obtained, the parameters R , L , and C for the bundle can be determined using the provided formula in the previous Sections 2 and 3. Note that $N_{CNT_tapered} < N_{CNT}$ is applied for the tapered via. One could notice that the major change after considering the tapered cylinder is the resistance and capacitance. Take an example of the CNTs bundle, the value of R_{CNTs} as described in Eq. 7 is increased due to the reduced N_{total} . The parasitic inductance of CNTs is supposed to be slightly reduced by the tapered structure but it can be safely assumed negligible, similar to the conclusion we observed in refs. [88–90].

The capacitances for the tapered via, as quoted from ref. [91] can be written as:

$$\frac{1}{C_{ox}} = \frac{1}{2\pi\epsilon_{ox}h_{via}} \sin\theta \ln \left(\frac{2r_{ox}'' + h_{via}\cot\theta}{2r_{via} + h_{via}\cot\theta} \right) \quad (41)$$

$$\frac{1}{C_{dep}} = \frac{1}{2\pi\epsilon_{Si}h_{via}} \sin\theta \ln \left(\frac{2r_{dep}'' + h_{via}\cot\theta}{2r_{ox}'' + h_{via}\cot\theta} \right) \quad (42)$$

where θ is the slope angle for the wall, r_{via}'' is the bottom radius of via, and r_{ox}'' is the bottom radius of the oxide liner. The total series capacitance p.u.l. is then $C'_{TSV} = (C_{ox}^{-1} + C_{dep}^{-1})^{-1}$ similar to the previous modeling of CNTs bundle in via by dropping the contribution of C_{CNTs} . Generally for the tapered via, the changes of capacitance for the CNTs TSV are not different from the conventional Cu TSV.

If we refer to the conventional case of Cu TSV, tapered via occasionally is intended to let the seed layer form a complete coverage inside the sidewall for conformal filling of Cu after the plating. If tapered via is not desired for CNTs growth, then the process development team knows how to fabricate a satisfyingly near-perfect cylindrical via by controlling the etch profiles of DRIE followed by wet etching of the scallops and grasses [92–94]. The process recipes for obtaining a perfectly cylindrical via and tapered via are not to be discussed further in this review.

4.2. Coaxial CNTs TSV

The concept of coaxial TSV for RF signal is similar to the coaxial cable with self-shielding and self-isolation functions to effectively control its characteristic impedance. Coaxial TSV consists of a central conductor, an insulating layer, a metallic shield, and an outer insulating layer as illustrated in Fig. 8a. Hence, it is an SG-type connection. There are several reports on the numerical modeling of the coaxial CNTs TSV but almost none on the actual fabrication. The configuration of coaxial CNTs TSV can be distinguished at least in two forms; i) heterogeneous coaxial where CNTs as the inner conductor with Cu ring shield, and ii) homogenous coaxial where CNTs as the inner conductor and ring shield.

Qu et al. provided the numerical modeling for the heterogeneous coaxial TSV with a configuration of CNTs(signal)-oxide-Si-oxide-Cu(shield)-oxide-Si(substrate) [95]. Fig. 8a shows the schematic of the proposed heterogeneous coaxial to indicate the radius of the components $r_1 - r_6$. The resistances of CNTs bundle and Cu, as quoted from ref. [95] can be written as:

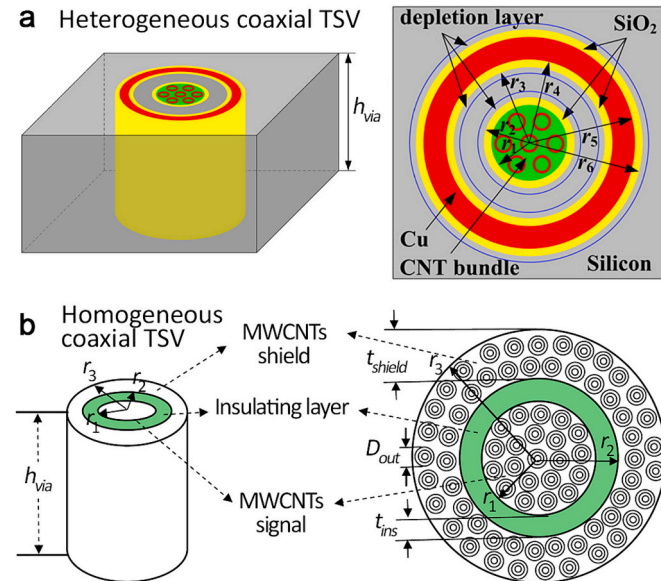


Fig. 8. a) Heterogeneous coaxial TSV with a configuration of CNTs(signal)-oxide-Si-oxide-Cu(shield)-oxide-Si(substrate), and b) homogenous coaxial TSV with a configuration of MWCNTs(signal)-insulator-MWCNTs(shield). It is more practical to fabricate heterogeneous coaxial TSV since the intermediate Si layer is required for forming oxide-Si-oxide isolation. Image a) is from ref. [95], and b) is from ref. [97].

$$R_{CNTs} = Re(Z_{CNTs}) \quad (43)$$

$$R_{Cu} = Re(Z_{Cu}) \quad (44)$$

The impedance of the coaxial TSV is related to the ratio of the electric and magnetic field amplitudes, where in ref. [95] both Z_{CNTs} and Z_{Cu} are solved using the Bessel functions of the first kind and the Bessel functions of the second kind as follows:

$$Z_{CNTs} = \frac{T_1 h_{via} I_0(T_1 r_1)}{2\pi r_1 \sigma_{CNTs} I_1(T_1 r_1)} \quad (45)$$

$$Z_{Cu} = \frac{T_2 h_{via} I_0(T_2 r_5) K_1(T_2 r_4) + I_1(T_2 r_4) K_0(T_2 r_5)}{\pi r_5 \sigma_{Cu} I_1(T_2 r_5) K_1(T_2 r_4) - I_1(T_2 r_4) K_1(T_2 r_5)} \quad (46)$$

where I is the modified Bessel functions of the first type ($I_0 = 0^{\text{th}}$ order, $I_1 = 1^{\text{st}}$ order), K is the modified Bessel functions of the second type ($K_0 = 0^{\text{th}}$ order, $K_1 = 1^{\text{st}}$ order), $T_1 = \sqrt{\omega\mu_0\sigma_{CNTs}}e^{j(\pi/4)}$, and $T_2 = \sqrt{\omega\mu_0\sigma_{Cu}}e^{j(\pi/4)}$, and $h_{via} = h_{CNTs} = h_{Cu}$ are assumed accordingly.

The internal inductances of CNTs bundle and Cu, and external inductance, as quoted from ref. [95] can be written as:

$$L_{CNTs} = Im(Z_{CNTs})/\omega \quad (47)$$

$$L_{Cu} = Im(Z_{Cu})/\omega \quad (48)$$

$$L_{ext} = \frac{\mu_0 h_{via}}{2\pi} \ln \frac{r_4}{r_1} \quad (49)$$

By ignoring the C_{CNTs} as we discussed earlier in Eq. 31, the capacitances between CNTs-Si and Cu-Si for this cylindrical coaxial via can be written as:

$$C_{CNTs-Si} = \left[\ln \frac{r_2 + t_{dep}}{r_2} + \frac{\ln \frac{r_2}{r_1}}{2\pi\epsilon_{Si}h_{via}} \right]^{-1} \quad (50)$$

$$C_{Cu-Si} = \left[\ln \frac{r_3}{r_3 - t_{dep}} + \frac{\ln \frac{r_3}{r_2}}{2\pi\epsilon_{ox}h_{via}} \right]^{-1} \quad (51)$$

Since there is conductive Si at the middle of layers CNTs-oxide-Si-oxide-Cu, it must be taken into account by including its capacitance and conductance:

$$C_{Si_mid} = \frac{2\pi\epsilon_{Si}h_{via}}{\ln(r_3/r_2)} \quad (52)$$

$$G_{Si_mid} = \frac{\sigma_{Si}C_{Si_mid}}{\epsilon_{Si}} = \frac{2\pi\sigma_{Si}h_{via}}{\ln(r_3/r_2)} \quad (53)$$

Note that these C_{Si_mid} and G_{Si_mid} are different to the external Si substrate as described in Eqs. 27 and 29, where the C_{Si} and G_{Si} are important for the RLGC model describing inter-TSVs interaction. Finally for this coaxial CNTs(signal)-oxide-Si-oxide-Cu(shield)-oxide-Si(substrate) TSV, the RLGC model for a single TSV without a neighbor can be written as:

$$R = R_{CNTs} + R_{Cu} \quad (54)$$

$$L = L_{CNTs} + L_{Cu} + L_{ext} \quad (55)$$

$$G_{coax} = \frac{\omega^2 C_1 G_{Si_mid}}{G_{Si_mid}^2 + \omega^2 (C_{Si_mid} + C_1)} \quad (56)$$

$$C = \frac{C_1 G_{Si_mid}^2 + \omega^2 C_1 C_{Si_mid} (C_{Si_mid} + C_1)}{G_{Si_mid}^2 + \omega^2 (C_{Si_mid} + C_1)^2} \quad (57)$$

where C_l is the series capacitance of the C_{CNTs_Si} and C_{Cu_Si} from the Eqs. 50 and 51, respectively. This exact $RLGC$ model by Qu et al. in ref. [95] is also employed in a recent conference paper to compare the insertion loss between this proposed structure of heterogeneous coaxial TSV with the coaxial Cu TSV [96].

The proposal of CNTs(signal)-oxide-Si-oxide-Cu(shield)-oxide-Si (substrate) as provided by Qu et al. in ref. [95] above, is quite complicated since there is a conductive Si middle layer between the signal and shield. There are several proposals with a simplified version with the configuration by omitting the Si middle layer and its oxide. Notably, we have a heterogeneous coaxial CNTs(signal)-oxide-Cu(shield) TSV modeled by Chandrakar et al. [88], and homogenous coaxial MWCNTs (signal)-insulator-MWCNTs(shield) TSV modeled by Su et al. in ref. [97] as illustrated in Fig. 8b. This kind of structure is easier to numerically model which is similar to the older proposal of coaxial Cu TSV [98–100]. However, this new proposal of MWCNTs-insulator-MWCNTs and the old proposal of Cu-oxide-Cu coaxial TSVs both share a similar major problem. They are almost impossible to fabricate since the Si middle layer is required to provide the base to isolate the signal and the shield. Fig. 9 is the proposed fabrication processes by Su et al. in ref. [97] for the MWCNTs(signal)-insulator-MWCNTs(shield) coaxial TSV. One could identify there are several issues during the fabrication process, especially on the transfer of MWCNTs(signal) from the second substrate to complete the coaxial structure.

4.3. Multi-bit CNTs TSV

The vertically aligned CNTs bundle is known to have anisotropic conduction for both heat and current. We observed a small research interest in numerical modeling of a novel structure multi-bit CNTs TSV. Theoretically, since the CNTs bundle does exhibit electrical anisotropies with the preferred current conduction in a vertical direction, we can build a multiple and independent conducting line inside the via by multiple contacts onto CNTs TSV. This configuration will allow the transmission of multiple independent signals and doubling the I/O counts without occupying additional areas on the chip (or interposer). The notable proposal for this technique can be found in a recent conference paper by Vaisband et al. in ref. [101]. Fig. 10a and b respectively show the schematic and equivalent circuit of the multi-bit CNTs-TSV, which is it can be either an SS-type of TSV (CM or DM signaling) or just an SG-type of TSV. In the paper, they described the bit resistance and mutual resistance for multi-bit CNTs TSV as:

$$R_{bit} = R_{CNTs} \cdot N_{bits} \quad (58)$$

$$R_m = \frac{1}{\sigma_{CNTs_lateral}} \frac{d_{12}}{2r_{via}h_{via}} N_{bits} \quad (59)$$

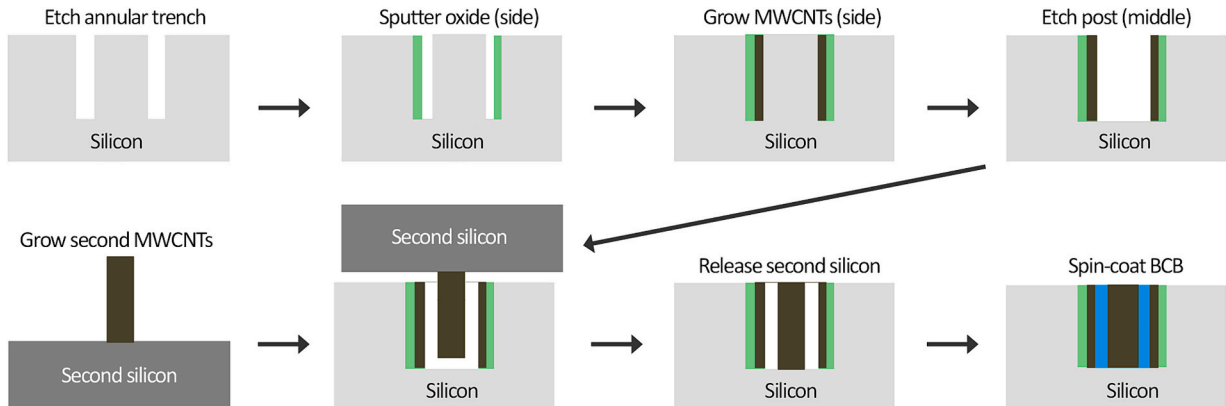


Fig. 9. One of the proposed fabrication processes of coaxial MWCNTs TSV without Si middle layer. Note that several shortcomings in this process cannot guarantee the successful fabrication of coaxial MWCNTs TSV, e.g. selective sidewall deposition of oxide and MWCNTs. The schematic is from ref. [97].

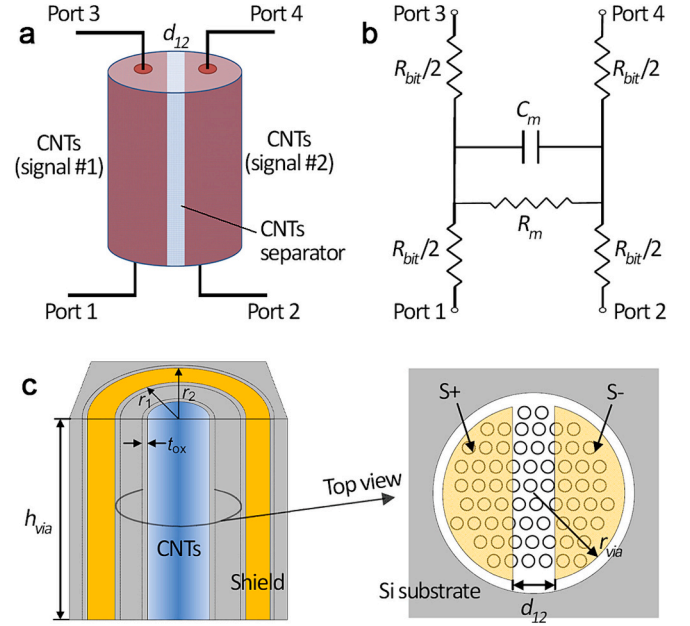


Fig. 10. a) Schematic of multi-bit CNTs TSV with CNTs separation layer, and b) equivalent circuit model for the multi-bit CNTs TSV with indicated mutual resistance and capacitance at the separation layer. c) Proposed structure of coaxial multi-bit CNTs TSV with DM signaling scheme. Images a,b) are from ref. [101], c) is from ref. [105].

where R_{CNTs} is the resistance of CNTs bundle in a cylindrical via, $\sigma_{CNTs_lateral}$ is the lateral conductivity of CNTs, d_{12} is the separation distance between two bits, and between $N_{bits} = 2$ is the number of independent signals propagating within the multi-bit CNTs TSV.

The proposal of multi-bit CNTs TSV by Vaisband et al. in ref. [101] above heavily depends on the validity of $\sigma_{CNTs_lateral}$. They take CNTs bundle as an anisotropic electrical conductor by $\sigma_{CNTs_lateral} = 10^{-7} \sigma_{CNTs}$. The definition of $\sigma_{CNTs_lateral}$ is not commonly investigated by the previous research and they quoted this empirical 10^{-7} from an experimental report by Zhu et al. in ref. [102]. In another experiment by Matsumoto et al. in ref. [103], the empirical can be adjusted to 10^{-4} between $\sigma_{CNTs_lateral}$ and σ_{CNTs} . As far as we are concerned, this definition of electrical anisotropic in the CNTs bundle stands on the assumption CNT strands are all in a vertical direction. The exact value of how much $\sigma_{CNTs_lateral}$ will be different with σ_{CNTs} , therefore, it greatly depends on the growth process of the CNTs bundle. It cannot promise a very good anisotropic characteristic for the CNTs with local and global twists.

Consider a multi-bit CNTs TSV with DM signal channels, then the

mutual capacitance between the bits can be written as:

$$C_m = \frac{\epsilon_{CNTs} A}{d_{12}} = \frac{2\epsilon_{CNTs} r_{via} h_{via} (N_{bits}/2)}{d_{12}} \quad (60)$$

where $A = 2r_{via}h_{via}$ is the cross-sectional area of the cylindrical via, and ϵ_{CNTs} is the permittivity of the CNTs bundle as the separation material between the two bits. At present, this proposal is still premature and the complete RLGC model of multi-bit CNTs TSV is not available in the literature. Note that L and C components depend on d_{12} , and it can be more complicated if via is tapered. The value of ϵ_{CNTs} needs to be determined either by experimental or simulation to prove that it can successfully be used to isolate two different signal lines.

Another notable work continuing investigating the multi-bits CNTs TSV can be found in refs. [104, 105] but for now we do not foresee this technology being viable soon. Hu et al. in a conference paper proposed the coaxial multi-bit CNTs TSV structure as illustrated in Fig. 10c [106]. This structure is not only challenging to fabricate at the part of multi-bit CNTs but also carries the same fabrication issue from the fabrication work of the annular Cu shield as discussed in the previous Section 4.2. In another example, Kumar et al. investigated the numerical modeling of multi-bit CNTs TGV and suggested the fabrication steps in Fig. 11 [107]. These steps, unfortunately, are not feasible to carry out due to the high-temperature growth of CNTs on glass and the challenge of forming the MWCNTs bundle at the periphery of the TGV.

4.4. Cu-CNTs composite TSV

The investigation of Cu-CNTs composite is an active research topic among material scientists for improving the mechanical strength and electrical conductivity of pure Cu. A notable report by Subramaniam et al. in ref. [108] measured a high ampacity (6×10^8 Acm⁻²) of Cu-CNTs composite which is 100 times higher than pure Cu. This merit is important for small features interconnect or TSV where the electromigration is the concerning issue. Commonly, Cu-CNTs composite can be formed by at least five methods; i) mixing of Cu nanoparticles and CNTs followed by sintering [109], ii) adding CNTs into Cu melt [110] iii) electrochemical deposition (ECD) using Cu source electrolyte mixed with CNTs dispersion [111], iv) physical vapor deposition of Cu on CNTs [112], and iv) ECD of Cu on CNTs [113]. Our literature study suggests the fabrication of Cu-CNTs TSV is feasible by ECD of Cu thick layer onto the CVD-grown vertically oriented CNTs. This process can be considered as a double work combining the process for CNTs TSV by CVD and Cu

TSV by ECD, which also can be translated into a doubled cost in fabrication. One of the earliest works of Cu-CNTs TSV was reported in a conference by Chai et al., where they successfully grow CNTs bundle inside a blind via and proceed to Cu ECD to form a partially completed TSV [114].

The process of Cu-CNTs TSV heavily relies on the superconformal ECD of Cu to fill the air void between the CNTs bundle. The formation of Cu filling inside the vacant areas of unfilled CNTs inside the via is purposely to improve the electrical conductivity of the TSV. An atomistic structural simulation of CNTs inside the Cu matrix suggests that Cu-CNTs composite is more resistant to electromigration compared to Cu due to a large Lorenz number of CNTs [115]. In one of the early reports on Cu-CNTs composite TSV by Feng and Burkett [116], they defined the Cu filling ratio inside the CNTs TSV as:

$$f_{Cu} = 1 - \frac{4N_{CNT}(d_{CNT} + \delta_{CNT})^2}{\pi d_{via}^2} \quad (61)$$

To be re-stated, N_{CNT} is the number of CNTs in a bundle within the cylindrical via (Eq. 7), d_{CNT} is the diameter of individual CNT strands, $\delta_{CNT} = 0.34$ nm is the Van der Waals intertube spacing, and d_{via} is the diameter of via. The conductivity of Cu-CNTs composite, as modified from ref. [117] can be written as:

$$\sigma_{Cu-CNTs} = f_{Cu}\sigma_{Cu} + (1 - f_{Cu})\sigma_{CNTs} \quad (62)$$

where the expression for the σ_{CNTs} is provided in Eq. 9. The temperature-dependent conductivity of Cu can be obtained from the empirical formula:

$$\sigma_{Cu} = \frac{5.814 \times 10^7}{1 + 3.9 \times 10^{-3}(T - 293 \text{ K})} \quad (63)$$

where the base conductivity of Cu is assumed to be 5.814×10^7 Sm⁻¹ at $T = 293$ K with the temperature coefficient of resistance of 3.9×10^{-3} K⁻¹. For the Cu TSV ($f_{Cu} = 1$), Fu et al. estimated its conductivity is equivalent to CNTs TSV ($f_{Cu} = 0$) with $F_m = 0.414$ [117]. In other words, theoretically all metallic CNTs TSV with $F_m = 1$ can surpass the performance of Cu TSV in terms of lower delay and power dissipation.

We learned the expectation $\sigma_{Cu-CNTs}$ from the numerical model may not be supported by experimental data. Feng and Burkett in another experimental investigation reported on the inferior $\sigma_{Cu-CNTs}$ compared with σ_{Cu} in the forms of TSV [118]. Their SEM images of CNTs TSV before and after Cu ECD are shown in Fig. 12a and b. Although they are

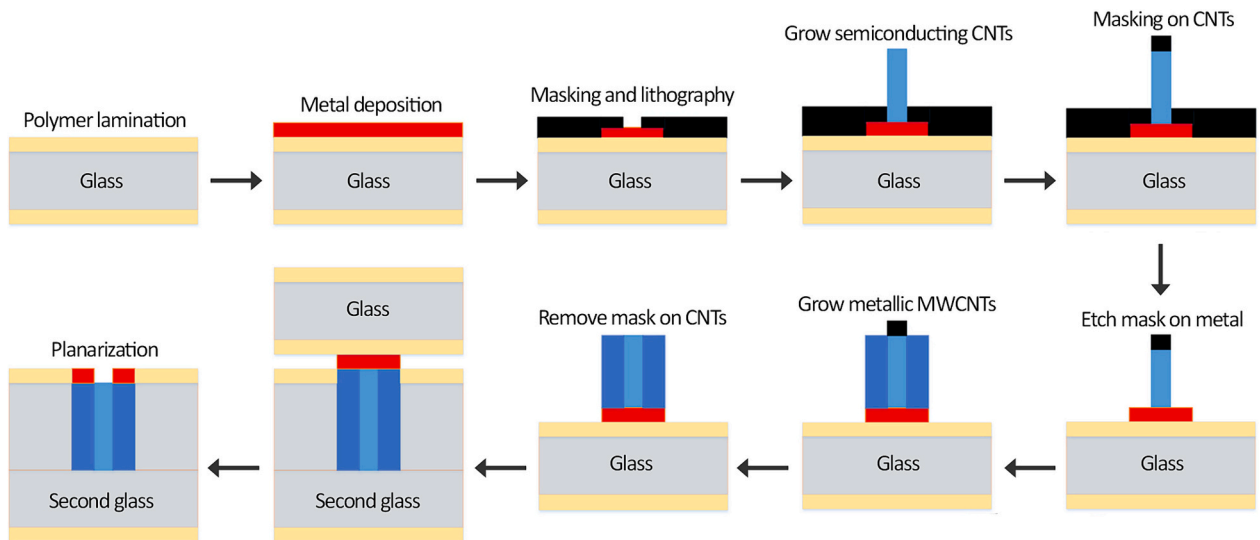


Fig. 11. One of the proposed fabrication processes of multi-bit CNTs TGV (using glass instead of Si). Note that several shortcomings in this process cannot guarantee the successful fabrication of multi-bit CNTs TGV, e.g. high-temperature growth of CNTs and MWCNTs on glass. The schematic is from ref. [107].

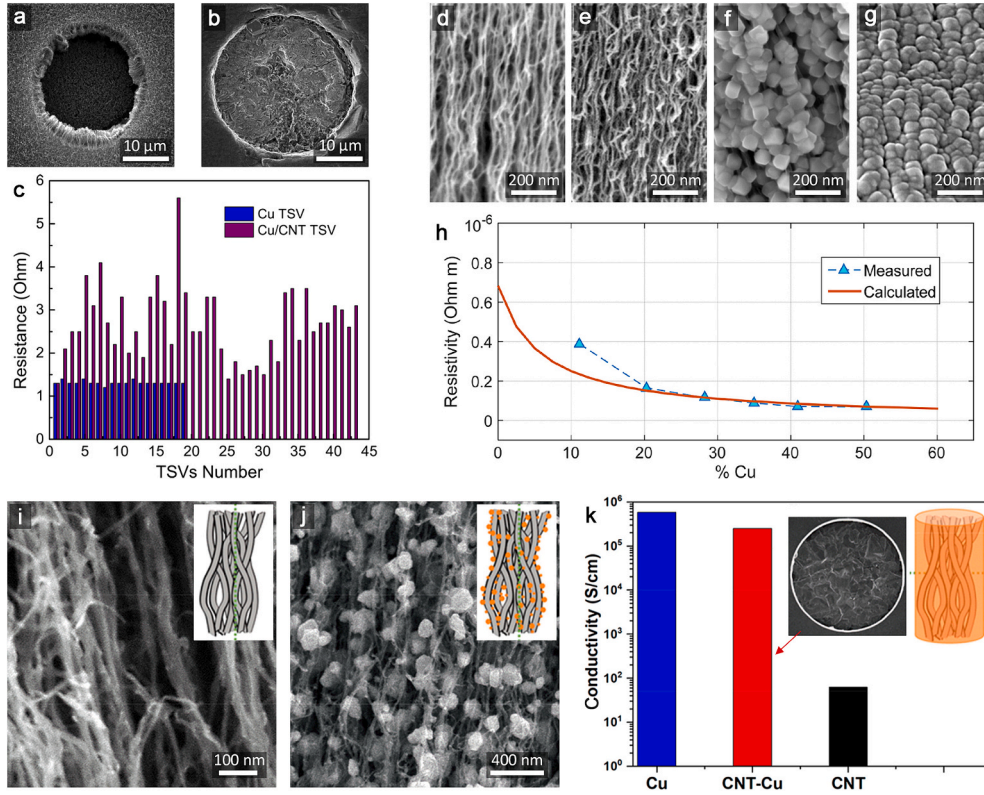


Fig. 12. First example of Cu-CNTs TSV: a) Si via after CVD growth of CNTs, b) Cu-CNTs after Cu ECD, and c) resistance of Cu-CNTs TSV versus Cu TSV. Second example: d–g) Formation of Cu-CNTs composite by ECD of Cu on CNTs, and h) resistivity of Cu-CNTs composite as a function of Cu volume. Third example: i,j) CNTs before and after Cu seeding, and k) conductivity of completed Cu-CNTs TSV compared with CNTs and Cu TSVs. Images a–c) are from ref. [118], d–h) are from ref. [119], and i–k) are from ref. [120].

quite satisfied with the Cu filling inside the CNTs bundle, the measured average $R_{Cu-CNTs} \sim 2.6 \Omega$ is double the value of $R_{Cu} \sim 1.3 \Omega$ as compared in Fig. 12c. These values can be converted into $\sigma_{Cu-CNTs} \sim 98 \text{ Scm}^{-1}$ and $\sigma_{Cu} \sim 196 \text{ Scm}^{-1}$ after considering the geometry of their via with $r_{via} = 12.5 \mu\text{m}$ and $h_{via} = 25 \mu\text{m}$. If we assume a good Cu filling into CNTs bundle with $f_{Cu} = 0.5$, then the original value σ_{CNTs} expected in Eq. 62 is then negligible compared with σ_{Cu} . In another experiment, Sun et al. improved the Cu ECD filling into the CNTs bundle by providing a Ti/Au seed layer onto the densified CNTs bundle to improve the wettability of CNTs and act as a conducting layer for the ECD process [119]. Figs. 12d – 12g provide the SEM images of the as-grown CNTs bundle, after the deposition of a Ti/Au seed layer, crystallization of Cu, and completed formation of Cu-CNTs composite, respectively. In Fig. 12h, they found Cu filling above 30% does not contribute to the lower resistivity of Cu-CNTs TSV. This also means Eq. 62 is only valid for the range of $f_{Cu} = 0 - 0.3$ (based on their observation).

A more recent report by Chen et al. validates Eq. 62 to be acceptable for determining the value of $\sigma_{Cu-CNTs}$ [120]. In the report, they explained the fabrication of Cu-CNTs TSV by a two-step Cu ECD process on a CNTs bundle. Figs. 12i and 2j show the SEM images of the CNTs bundle before and after being seeded by Cu using the first step ECD process. Later on with this uniformly seeded Cu nanoparticles, they conducted the second step of the ECD process to deposit thick Cu filling the void inside the CNTs bundle with estimated $f_{Cu} = 0.5 - 0.65$. The measured $\sigma_{Cu-CNTs} \sim 2.5 \times 10^5 \text{ Scm}^{-1}$ is significantly higher than $\sigma_{CNTs} \sim 60 \text{ Scm}^{-1}$ ($f_{Cu} = 0$), but still lower than $\sigma_{Cu} \sim 5.8 \times 10^5 \text{ Scm}^{-1}$ as compared in Fig. 12k. Using Eq. 62, the Cu-CNTs with $f_{Cu} = 0.5 - 0.65$ should have $\sigma_{Cu-CNTs} = (2.9 - 3.8) \times 10^5 \text{ Scm}^{-1}$ which is not too far from their measurement. Since now we have one proof that Eq. 62 is valid, we can treat the previous findings by Feng and Burkett in ref. [118] and Sun et al. in ref. [119] as deviation from a desired condition of Cu-CNTs composite. For

now, we cannot point out a specific reason for this observation but it is most likely related to how well Cu fills the air void inside CNTs bundle and how electrons are transported within the composite.

With the verification of the relationship between $\sigma_{Cu-CNTs}$ and f_{Cu} by experimental, the understanding of Cu-CNTs TSV can be continued with numerical modeling and simulation. We observed the numerical modeling of Cu-CNTs TSV mostly revolves around the problem of the skin effect of the conductor inside the via. Consider a cylindrical via filled with a Cu-CNTs composite, the distribution of current density at the cross-section of via as quoted from ref. [117], can be obtained from the following differential equation:

$$\frac{d^2 J}{dr^2} + \frac{dJ}{rdr} + K^2 J = 0 \quad (64)$$

where J is the current density, r is the radial direction, and we define $K = \sqrt{-j\omega\mu_0\sigma_{Cu-CNTs}}$. Therefore, the current density along the radial direction can be written as:

$$J(r) = \frac{KI}{2\pi r_0} \times \frac{J_0(Kr)}{J_1(Kr_0)} \quad (65)$$

where I is the current flowing inside the TSV, and J is the Bessel functions of the first type ($J_0 = 0^{\text{th}}$ order, $J_1 = 1^{\text{st}}$ order). Similar to Section 3.2, the skin depth in Cu-CNTs TSV is given by $\delta_{TSV} = \sqrt{2/\omega\mu_0\sigma_{Cu-CNTs}}$. Note that $\sigma_{Cu-CNTs}$ (similarly for the σ_{Cu} and σ_{CNTs}) is the function of frequency. In Eq. 4, the kinetic inductance p.u.l. of CNT is defined as $L_K = h/(4e^2 v_F) \approx 16.1 \text{ nH } \mu\text{m}^{-1}$, hence CNT itself is not sensitive to the skin effect because this L_K is a frequency-independent parameter.

Fig. 13a shows the schematic of Cu filling inside the CNTs bundle in a simulation work by Feng and Burkett [116]. They build the RL model for the Cu-CNTs composite, which R_{Cu} is parallel with Z_{CNTs} as indicated in

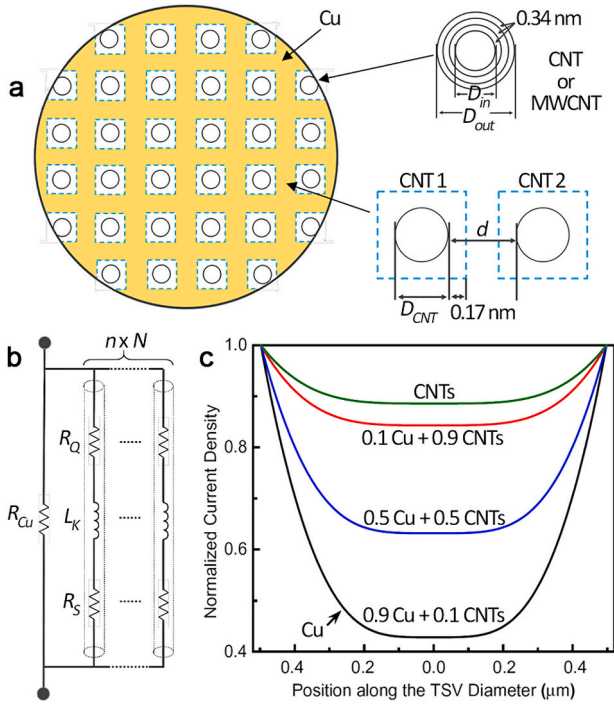


Fig. 13. a) Schematic top view of Cu-CNTs TSV indicated with the location and filling of Cu between CNT strands, b) equivalent circuit model of Cu-CNTs, and c) normalized current density versus position along TSV diameter for the varied Cu filling in the composite. Images are from ref. [116].

Fig. 13b. Using Eqs. 64 and 65, they show how increasing f_{Cu} results in lowered current density at the center of TSV due to the skin effect. Fig. 13c is the comparison between CNTs and Cu-CNTs TSV at a frequency of 100 GHz with a flowing current of 0.1 mA. Therefore, Cu-CNTs with a moderate value of f_{Cu} is better compared with Cu due to the significant reduction in skin effect. Similarly, Zheng et al. in ref. [55] concludes Cu-CNTs are better compared with Cu, provided that the value of f_{Cu} is lowered and F_m is increased.

Rao uses this concept and RL analysis for the mixed bundle of CNTs-MWCNTs forming the composite with Cu inside the tapered via [121,122]. The author reaches the same conclusion that $f_{Cu} \leq 0.8$ should be optimum to avoid Cu dominating the unequal current distribution due to the skin effect at high operational frequency. Additionally, for the Cu-MWCNTs-CNTs composite TSV, the author found the mix of 50% CNTs and 50% MWCNTs with $f_{Cu} = 0.3$ results in similar S_{11} and S_{21} parameters to the Cu TSV. Higher f_{Cu} is not preferable as it will reduce the conductivity of the CNTs-MWCNTs mix bundle and result in higher delay. The comprehensive RLGC model for the Cu-CNTs TSV accounting for the changes in ballistic resistance and inductance after Cu filling, however, is still not available in the literature. Apart from TL modeling, Nadir et al. opted for ESC modeling for the two Cu-CNTs interconnects by specifying the values of R , L , and C to conduct a circuit simulation [123].

We see a potential in Cu-CNTs TSV technology but for now, the anticipated fabrication process is limited to CVD growth of CNTs bundle followed by Cu ECD. It is understandable since only CVD-grown CNT can yield to the vertically aligned structure which is the whole concept of CNTs TSV begin with to fill the via. If there is any insight or preliminary works in the future that can fabricate and test the Cu-CNTs TSV by different fabrication approaches (e.g. ECD using Cu source electrolyte mixed with CNTs dispersion into blind via, or sintering the mixed powder of Cu nanoparticles inside the via), then we can further evaluate the investment of this Cu-CNTs TSV from an economical perspective. Using the alternative approaches that exclude the need for CVD, CNTs will not be in the form of a vertically aligned structure but they will

randomly fill inside the Cu matrix. When this happens, the RLGC modeling works of 1D CNTs as described in previous sections are no longer relevant and researchers may treat this material simply as a modified Cu. The main characteristics of Cu-CNTs as a conductor, e.g. electrical conductivity and thermal conductivity should be redefined based on the volume fraction of both Cu and CNTs within the Cu-CNTs composite. Researchers can simply feed this material information into a 3D model using commercial software, e.g. Ansys to generate the RLGC of this Cu-CNTs composite, and ultimately conduct a fully automated analysis (see the workflow in the previous Fig. 7).

5. Summary and conclusion

The understanding of CNTs as a conductor in TSV mostly relies on numerical modeling. Although several works focus on the experiment, the offered information is only on the resistance of the fabricated CNTs TSV. One of the big concerns that we can summarize from this review is whether the measured resistance from the experimental works matches the value of R as in the RLC and RLGC models. A straightforward answer is no because the progress in numerical modeling of CNTs TSV is heavily invested in theoretically compact packing of CNTs in a bundle with Van der Waals spacing of only 0.34 nm. We perceived that recent advancements in CVD growth techniques for high-density CNT forests are barely $\sim 1\%$ of the theoretically compact packing. For a single TL problem, all R , L , and C parameters of CNTs depend on N_{CNT} which is the number of CNT strands in the bundle. The large theoretical value of N_{CNT} introduces errors in the RLC model. Hence, it will not represent the actual behavior of the fabricated CNTs TSV during the transmission of digital or analog signals. Since the GC component is determined by Si substrate and oxide liner, the RLGC modeling for the simple cylindrical, tapered, coaxial, multi-bit, and composite CNTs TSVs will also be flawed due to the overestimation of N_{CNT} . It will not represent the actual behavior of the fabricated CNTs TSVs during the operation neither in 3D IC nor 2.5D packaging.

This review is uncertain about the direction of CNTs TSV replacing Cu TSV for high-frequency RF applications, specifically at 100 GHz and above. CNTs are indeed less susceptible to skin effect compared to Cu but these frequency regions are actually more concerned with the Si substrate itself rather than the conductor. In other words, targets for the CNTs TGV and Cu-CNTs TGV are more sensible than CNTs TSV. Replacing the Si substrate with glass also means this topic is specific to the 2.5D packaging of RF ICs. Thus, the numerical modeling supposedly needs to be aware of the geometry of TGV for package-level (e.g. diameter, aspect ratio, and pitch), and simulation should not emphasize digital indicators (e.g. delay and power dissipation). Advanced topics that are related when considering CNTs TGV for 2.5D packaging of RF ICs are: i) tapered via, ii) coaxial, and iii) Cu-CNTs composite including the consideration of differential signaling. The RLGC modeling needs to have a correction for the N_{CNT} of CNTs bundle inside the TGV.

For the high-speed digital 3D IC, we hardly see CNTs TSV as a replacement for Cu TSV. Related to the problem of N_{CNT} above, the resistances of CNTs TSV and Cu-CNTs are many times measured higher compared to the Cu TSV in independent experimental reports. As a consequence, researchers cannot truly anticipate the actual performance of CNTs TSV in 3D IC since the RC model always overestimates its performance against Cu TSV. The motivation for multi-bit CNTs TSV for 3D IC at this point can be safely discarded after considering the factor above and the unproven electrical anisotropies of CNTs that can sustain signal separation within a single TSV. For the high-speed digital 2.5D packaging, CNTs TSV is also not seen as a replacement for Cu TSV as the typical digital frequency is not that high for the Cu skin effect. The large size of package-level Cu TSV makes it less susceptible to electro-migration, thus invalidating the higher ampacity of CNTs over Cu. We expect the trend of Cu and Cu-CNTs TGVs (replace Si with glass) will continue to grow instead of replacing Cu itself with CNTs for 2.5D packaging of both digital and analog ICs.

In conclusion, although there are some flaws, the progress of numerical modeling does help determine the feasibility and direction of this CNTs TSV technology. Future works need to be aware of the end application of CNTs TSV, thus they can focus on answering the digital and analog requirements either for 3D IC or 2.5D packaging. Recommendations summarized based on our review can help future works to not focus on the dead-end paths where CNTs do not hold much promise.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

- [1] G.E. Moore, Cramming more components onto integrated circuits, *Electronics* 38 (1965) 114–117.
- [2] Executive summary, International Technology Roadmap for Semiconductors, 2005. Edition.
- [3] J.H. Lau, *Semiconductor Advanced Packaging*, Springer Nature, 2021.
- [4] R.K. Ratnesh, A. Goel, G. Kaushik, H. Garg, M. Chandan, B. Prasad Singh, Advancement and challenges in MOSFET scaling, *Mater. Sci. Semicond. Process.* 134 (2021) 106002.
- [5] Y. Ryu, S.G. Ahn, J.H. Lee, J. Park, Y.K. Kim, H. Kim, Y.G. Song, H.W. Cho, S. Cho, S.H. Song, H. Lee, U. Shin, J. Ahn, J.M. Ryu, S. Lee, K.H. Lim, J. Lee, J. H. Park, J.S. Jeong, S. Joo, D. Cho, S.Y. Kim, M. Lee, H. Kim, M. Kim, J.S. Kim, J. Kim, H.G. Kang, M.K. Lee, S.R. Kim, Y.C. Kwon, Y.Y. Byun, K. Lee, S. Park, J. Youn, M.O. Kim, K. Sohn, S.J. Hwang, J. Lee, A 16 GB 1024 GB/s HBM3 DRAM with source-synchronized bus design and on-die error control scheme for enhanced RAS features, *IEEE J. Solid State Circuits* 58 (2023) 1051–1061.
- [6] J.P. Gambino, S.A. Adderly, J.U. Knickerbocker, An overview of through-silicon-via technology and manufacturing challenges, *Microelectron. Eng.* 135 (2015) 73–106.
- [7] B. Song, Z. Li, X. Wang, X. Fu, F. Liu, L. Jin, Z. Huo, Analysis of HBM failure in 3D NAND flash memory, *Electronics* 11 (2022) 944.
- [8] T.H. Hung, Y.M. Pan, K.N. Chen, Stress issue of vertical connections in 3D integration for high-bandwidth memory applications, *Memories – Mater. Devices Circuits Syst.* 4 (2023) 100024.
- [9] J. Wu, S. Naffziger, P.J. Shyvers, M.S. Bhagavat, K. Mysore, B.P. Wilkerson, Arrangement and thermal management of 3D stacked dies, US Patent Application Publication, US2019/0067152A1, 2019.
- [10] S. Chang, H. Shin, 2.4-GHz CMOS Bluetooth RF receiver with improved IM2 distortion tolerance, *IEEE Trans. Microw. Theory Tech.* 68 (2020) 4589–4598.
- [11] J. Chen, M. Berg, K. Rasilainen, Z. Siddiqui, M.E. Leinonen, A. Parssinen, Broadband cross-slotted patch antenna for 5G millimeter-wave applications based on characteristic mode analysis, *IEEE Trans. Antennas Propag.* 70 (2022) 11277–11292.
- [12] X. Yi, G. Feng, Z. Liang, C. Wang, B. Liu, C. Li, K. Yang, C.C. Boon, Q. Xue, A 24/77 GHz dual-band receiver for automotive radar applications, *IEEE Access* 7 (2019) 48053–48059.
- [13] J.O. Plouchart, X. Gu, W. Lee, A. Tzadok, D. Liu, H. Liu, M. Yeck, C. Baks, A. V. Garcia, Si-based 94-GHz phased array transmit and receive modules for real-time 3D radar imaging, in: 2019 IEEE MTT-S International Microwave Symposium, 2019, pp. 532–535.
- [14] H. Wang, J. Ma, Y. Yang, M. Gong, Q. Wang, A review of system-in-package technologies: application and reliability of advanced packaging, *Micromachines* 14 (2023) 1149.
- [15] V.R. Kumar, M.K. Majumder, B.K. Kaushik, Graphene-based on-chip interconnects and TSVs, *IEEE Nanotechnol. Mag.* 8 (2014) 14–20.
- [16] G. Mokry, J. Pozuelo, J.J. Vilatela, J. Sanz, J. Baselga, High ampacity carbon nanotube materials, *Nanomaterials* 9 (2019) 383.
- [17] Z. Han, A. Fina, Thermal conductivity of carbon nanotubes and their polymer nanocomposites: a review, *Prog. Polym. Sci.* 36 (2011) 914–944.
- [18] S.L. Burkett, M.B. Jordan, R.P. Schmitt, L.A. Menk, A.E. Hollowell, Tutorial on forming through-silicon vias, *J. Vac. Sci. Technol. A* 38 (2020) 031202.
- [19] M. Kumar, Y. Ando, Chemical vapor deposition of carbon nanotubes: a review on growth mechanism and mass production, *J. Nanosci. Nanotechnol.* 10 (2010) 3739–3758.
- [20] G. Chen, D.N. Futaba, K. Hata, Catalysts for the growth of carbon nanotube “forests” and superaligned arrays, *MRS Bull.* 42 (2017) 802–808.
- [21] K.M. Vanderburgh, S.J. Park, F. Fornasiero, Growth of carbon nanotube forests on flexible metal substrates: advances, challenges, and applications, *Carbon* 206 (2023) 402–421.
- [22] T. Wang, K. Jeppson, N. Olofsson, E.E.B. Campbell, J. Liu, Through silicon vias filled with planarized carbon nanotube bundles, *Nanotechnology* 20 (2009) 485203.
- [23] D. Jiang, W. Mu, S. Chen, Y. Fu, K. Jeppson, J. Liu, Vertically stacked carbon nanotube-based interconnects for through silicon via application, *IEEE Electron Device Lett.* 36 (2015) 499–501.
- [24] T. Goyal, M.K. Majumder, B.K. Kaushik, Modeling and fabrication aspects of cu-and carbon nanotube-based through-silicon vias, *IETE J. Res.* 67 (2021) 377–393.
- [25] M. Vizuete, M.J. Gomez-Escalonilla, S. Garcia-Rodriguez, J.L.G. Fierro, P. Atienzar, H. Garcia, F. Langa, Photochemical evidence of electronic interwall communication in double-wall carbon nanotubes, *Chem. Eur. J.* 18 (2012) 16922–16930.
- [26] P.L. McEuen, M.S. Fuhrer, H. Park, Single-walled carbon nanotube electronics, *IEEE Trans. Nanotechnol.* 1 (2002) 78–85.
- [27] N. Srivastava, R.V. Joshi, K. Banerjee, Carbon nanotube interconnects: implications for performance, power dissipation and thermal management, *IEEE Int. Electron Devices Meet.* (2005) 249–252.
- [28] N. Srivastava, K. Banerjee, Performance analysis of carbon nanotube interconnects for VLSI applications, *IEEE/ACM Int. Conf. Comp. Aided Design* (2005) 383–390.
- [29] S. Ryu, R. Lopez, L. Serra, Conductance of electrostatic wire junctions in bilayer graphene, *Phys. Rev. B* 106 (2022) 035424.
- [30] J.Y. Park, S. Rosenblatt, Y. Yaish, V. Sazonova, H. Ustunel, S. Braig, T.A. Arias, P. W. Brouwer, P.L. McEuen, Electron–phonon scattering in metallic single-walled carbon nanotubes, *Nano Lett.* 4 (2004) 517–520.
- [31] S. Kannan, B. Kim, A. Gupta, S.H. Noh, L. Li, Characterization of high performance CNT-based TSV for high-frequency RF applications, *Adv. Mater. Res.* 1 (2012) 37–49.
- [32] N. Chiodarelli, K. Kellens, D.J. Cott, N. Peys, K. Arstila, M. Heyns, S.D. Gendt, G. Groeseneken, P.M. Vereecken, Integration of vertical carbon nanotube bundles for interconnects, *J. Electrochem. Soc.* 157 (2010) K211–K217.
- [33] Y. Matsuda, W.Q. Deng, W.A. Goddard III, Contact resistance for “end-contacted” metal-graphene and metal-nanotube interfaces from quantum mechanics, *J. Phys. Chem. C* 114 (2010) 17845–17850.
- [34] T. Goyal, M.K. Majumder, B.K. Kaushik, Propagation delay and power dissipation for different aspect ratio of single-walled carbon nanotube bundled TSV, *J. Semicond.* 36 (2015) 065001.
- [35] A. Srivastava, Y. Xu, A.K. Sharma, Carbon nanotubes for next generation very large scale integration interconnects, *J. Nanophoton.* 4 (2010) 041690.
- [36] J. Dai, J. Li, H. Zeng, X. Cui, Measurements on quantum capacitance of individual single walled carbon nanotubes, *Appl. Phys. Lett.* 94 (2009) 093114.
- [37] B. Xu, R. Chen, J. Zhou, J. Liang, Recent progress and challenges regarding carbon nanotube on-chip interconnects, *Micromachines* 13 (2022) 1148.
- [38] A.V. Emelianov, N.P. Nekrasov, M.V. Moskotin, G.E. Fedorov, N. Otero, P. M. Romero, V.K. Nevolin, B.I. Afonin, A.G. Nasibulin, I.I. Bobrinetskiy, Individual SWCNT transistor with photosensitive planar junction induced by two-photon oxidation, *Adv. Electron. Mater.* 7 (2021) 2000872.
- [39] A. Alam, M.K. Majumder, A. Kumari, V.R. Kumar, B.K. Kaushik, Performance analysis of single- and multi-walled carbon nanotube based through silicon vias, in: 2015 IEEE 65th Electronic Components and Technology Conference, 2015, pp. 1834–1839.
- [40] A.K. Kureshi, M. Hasan, Analysis of CNT bundle and its comparison with copper interconnect for CMOS and CNFET drivers, *J. Nanomater.* 2009 (2009) 486979.
- [41] F. Kreupl, A.P. Graham, G.S. Duesberg, W. Steinhogel, M. Liebau, E. Unger, W. Honlein, Carbon nanotubes in interconnect applications, *Microelectron. Eng.* 64 (2002) 399–408.
- [42] N.T. Dee, M. Schneider, D.N. Zakharov, P.R. Kidambi, A.J. Hart, Automated processing of environmental transmission electron microscopy images for quantification of thin film dewetting and carbon nanotube nucleation dynamics, *Carbon* 192 (2022) 249–258.
- [43] T. Xu, Z. Wang, J. Miao, X. Chen, C.M. Tan, Aligned carbon nanotubes for through-wafer interconnects, *Appl. Phys. Lett.* 91 (2007) 042108.
- [44] S. Choudhary, N. Singh, P. Ranjan, Induced magnetization in armchair and zigzag CNTs on adsorbing transition metals, *J. Magn. Magn. Mater.* 538 (2021) 168287.
- [45] M. Amekpewu, S.Y. Mensah, R. Musah, S.S. Abukari, N.G. Mensah, K.A. Dompheh, Radius dependence of the electrical conductivity of zigzag carbon nanotubes, *Phys. E: Low-Dimens. Syst. Nanostruct.* 130 (2021) 114712.
- [46] S. Ilani, P.L. McEuen, Electron transport in carbon nanotubes, *Annu. Rev. Condens. Matter Phys.* 1 (2010) 1–25.
- [47] M.V. Kharlamova, Investigation of growth dynamics of carbon nanotubes, *Beilstein J. Nanotechnol.* 8 (2017) 826–856.
- [48] T. Wang, K. Jeppson, L. Ye, J. Liu, Carbon-nanotube through-silicon via interconnects for three-dimensional integration, *Small* 22 (2011) 2313–2317.
- [49] Z. Yu, P.J. Burke, Microwave transport in metallic single-walled carbon nanotubes, *Nano Lett.* 5 (2005) 1403–1406.
- [50] W. Wang, S. Haruehanroengra, L. Shang, M. Liu, Inductance of mixed carbon nanotube bundles, *Micro, Nano Lett.* 2 (2007) 35–39.

- [51] M.K. Majumder, B.K. Kaushik, S.K. Manhas, Analysis of delay and dynamic crosstalk in bundled carbon nanotube interconnects, *IEEE Trans. Electromagn. Compat.* 56 (2014) 1666–1673.
- [52] W.S. Zhao, L. Sun, W.Y. Yin, Y.X. Guo, Electrothermal modelling and characterisation of submicron through-silicon carbon nanotube bundle vias for three-dimensional ICs, *Micro Nano Lett.* 9 (2014) 123–126.
- [53] L. Qian, Y. Xia, G. Shi, J. Wang, Y. Ye, S. Du, Electrical-thermal characterization of through packaging vias in glass interposer, *IEEE Trans. Nanotechnol.* 16 (2017) 901–908.
- [54] B. Vaisband, A. Maurice, C.W. Tan, B.K. Tay, E.G. Friedman, Electrical and thermal models of CNT TSV and graphite interface, *IEEE Trans. Electron Dev.* 65 (2018) 1880–1886.
- [55] J. Zheng, Z.Q. Su, G.Y. Wang, M. Li, W.S. Zhao, G. Wang, Circuit modeling of Cu/CNT composite through-silicon vias (TSV), in: 2015 IEEE MTT-S International Microwave Workshop Series on Advanced Materials and Processes for RF and THz Applications, 2015, pp. 1–3.
- [56] S. Subash, M.H. Chowdhury, Mixed carbon nanotube bundles for interconnect applications, *Int. J. Electron.* 96 (2009) 657–671.
- [57] A. Naemi, J.D. Meindl, Physical modeling of temperature coefficient of resistance for single- and multi-wall carbon nanotube interconnects, *IEEE Electron Device Lett.* 28 (2007) 135–138.
- [58] X. Li, A. Cao, Y.J. Jung, R. Vajtai, P.M. Ajayan, Bottom-up growth of carbon nanotube multilayers: unprecedented growth, *Nano Lett.* 5 (2005) 1997–2000.
- [59] K. Ghosh, N. Ranjan, Y.K. Verma, C.S. Tan, Graphene-CNT hetero-structure for next generation interconnects, *RSC Adv.* 6 (2016) 53054–53061.
- [60] H. Fiedler, M. Toader, S. Hermann, M. Rennau, R.D. Rodriguez, E. Sheremet, M. Hietschold, D.R.T. Zahn, S.E. Schulz, T. Gessner, Back-end-of-line compatible contact materials for carbon nanotube based interconnects, *Microelectron. Eng.* 137 (2015) 130–134.
- [61] R. Xie, C. Zhang, M.H. van der Veen, K. Arstila, T. Hantschel, B. Chen, G. Zhong, J. Robertson, Carbon nanotube growth for through silicon via application, *Nanotechnology* 24 (2013) 125603.
- [62] M.D. Volder, S.H. Tawfik, S.J. Park, D. Copic, Z. Zhao, W. Lu, A.J. Hart, Diverse 3D microarchitectures made by capillary forming of carbon nanotubes, *Adv. Mater.* 22 (2010) 4384–4389.
- [63] T. Wang, S. Chen, D. Jiang, Y. Fu, K. Jeppson, L. Ye, J. Liu, Through-silicon vias filled with densified and transferred carbon nanotube forests, *IEEE Electron Device Lett.* 33 (2012) 420–422.
- [64] H. Li, N. Srivastava, J.F. Mao, W.Y. Yin, K. Banerjee, Carbon nanotube vias: does ballistic electron–phonon transport imply improved performance and reliability? *IEEE Trans. Electron Dev.* 58 (2011) 2689–2701.
- [65] H. Li, W.Y. Yin, K. Banerjee, J.F. Mao, Circuit modeling and performance analysis of multi-walled carbon nanotube interconnects, *IEEE Trans. Electron Dev.* 55 (2008) 1328–1337.
- [66] M.S. Sarto, A. Tamburrano, Single-conductor transmission-line model of multiwall carbon nanotubes, *IEEE Trans. Nanotechnol.* 9 (2010) 82–92.
- [67] V.R. Kumar, B.K. Kaushik, A. Patnaik, Crosstalk noise modeling of multiwall carbon nanotube (MWCNT) interconnects using finite-difference time-domain (FDTD) technique, *Microelectron. Reliab.* 55 (2015) 155–163.
- [68] M. D'Amore, M.S. Sarto, A. Tamburrano, Fast transient analysis of next-generation interconnects based on carbon nanotubes, *IEEE Trans. Electromagn. Compat.* 52 (2010) 496–503.
- [69] M. Tang, J. Lu, J. Mao, Study on equivalent single conductor model of multi-walled carbon nanotube interconnects, in: 2012 Asia Pacific Microwave Conference Proceedings, 2012, pp. 1247–1249.
- [70] B.K. Kaushik, V.R. Kumar, M.K. Majumder, A. Alam, Modeling and Performance Analysis of CNT-Based through Silicon Vias, CRC Press, Through Silicon Vias, 2016, pp. 97–124.
- [71] W.S. Zhao, W.Y. Yin, Y.X. Guo, Electromagnetic compatibility-oriented study on through silicon single-walled carbon nanotube bundle via (TS-WCNTBV) arrays, *IEEE Trans. Electromagn. Compat.* 54 (2012) 149–157.
- [72] A.E. Engin, S.R. Narasimhan, Modeling of crosstalk in through silicon vias, *IEEE Trans. Electromagn. Compat.* 55 (2013) 149–158.
- [73] W.S. Zhao, W.Y. Yin, X.P. Wang, X.L. Xu, Frequency- and temperature-dependent modeling of coaxial through-silicon vias for 3-D ICs, *IEEE Trans. Electron Dev.* 58 (2011) 3358–3368.
- [74] C.C. Sahu, S. Anand, M.K. Majumder, An analysis of the eddy effect in through-silicon vias based on Cu and CNT bundles: the impact on crosstalk and power, *J. Comput. Electron.* 20 (2021) 2456–2470.
- [75] J. Kim, I. Hwang, Y. Kim, J. Cho, V. Sundaram, R. Tummala, J. Kim, Precise RLGC modeling and analysis of through glass via (TGV) for 2.5D/3D IC, in: 2015 IEEE 65th Electronic Components and Technology Conference, 2015, pp. 254–259.
- [76] L. Qian, G. Shi, Y. Ye, Electrical-thermal characterization of carbon nanotube based through glass vias for 3-D integration, in: 2016 IEEE 16th International Conference on Nanotechnology, 2016, pp. 663–666.
- [77] Y. Liu, Z. Zhu, X. Liu, Q. Lu, X. Yin, L. Guo, Y. Yang, Electromagnetic modeling and analysis of the tapered differential through glass vias, *Microelectron. J.* 83 (2019) 27–31.
- [78] C. Xu, H. Li, R. Suaya, K. Banerjee, Compact AC modeling and analysis of Cu, W, and CNT based through-silicon vias (TSVs) in 3-D ICs, in: 2009 IEEE International Electron Devices Meeting, 2009, pp. 1–4.
- [79] C. Xu, H. Li, R. Suaya, K. Banerjee, Compact AC modeling and performance analysis of through-silicon vias in 3-D ICs, *IEEE Trans. Electron Dev.* 57 (2010) 3405–3417.
- [80] S. Chandrakar, D. Gupta, M.K. Majumder, Impact of TSV bump and redistribution layer on crosstalk delay and power loss, *memories – mater. Devices Circuits Syst.* 4 (2023) 100040.
- [81] L. Qian, Y. Xia, G. Shi, Electrical modeling and analysis of a mixed carbon nanotube based differential through silicon via in 3-D integration, *IEEE Trans. Nanotechnol.* 15 (2016) 155–163.
- [82] J.W. Pan, K. Fu, Z.H. Cheng, W.S. Zhao, G. Wang, Modeling of crosstalk effects in carbon nanotube based differential through-silicon via array, in: 2017 IEEE Electrical Design of Advanced Packaging and Systems Symposium, 2017, pp. 1–3.
- [83] W.S. Zhao, J. Zheng, F. Liang, K. Xu, X. Chen, G. Wang, Wideband modeling and characterization of differential through-silicon vias for 3-D ICs, *IEEE Trans. Electron Dev.* 63 (2016) 1168–1175.
- [84] J.W. Pan, K. Fu, Q. Liu, W.S. Zhao, L. Dong, G. Wang, Modelling of crosstalk in differential through silicon vias for three-dimensional integrated circuits, *IET Microwaves Antennas Propag.* 13 (2019) 1529–1535.
- [85] Y. Sun, J. Luo, G. Ding, Modeling and analysis of TSV arrays with different ground and signal distributions in 2.5D/3D integration systems, *J. Phys. Conf. Ser.* 1087 (2018) 052019.
- [86] S.J. Bleiker, A.C. Fischer, U. Shah, N. Somjit, T. Haraldsson, N. Roxhed, J. Oberhammer, G. Stemme, F. Niklaus, High-aspect-ratio through silicon vias (TSVs) for high-frequency application fabricated by magnetic assembly of gold-coated nickel wires, *IEEE Trans. Compon. Pack. Manuf. Technol.* 5 (2015) 21–27.
- [87] C.H. Wu, L. Shen, H. Zhang, J. Yan, D.J. Hou, G. Zhou, Y.L. Wu, Equivalent circuit parameters of planar transmission lines with spoof surface plasmon polaritons and its application in high density circuits, *Sci. Rep.* 9 (2019) 18853.
- [88] S. Chandrakar, D. Gupta, M.K. Majumder, Performance analysis of Cu/CNT based TSV: impact on crosstalk and power, *J. Comput. Electron.* 21 (2022) 1262–1274.
- [89] Y. Liang, Y. Li, Closed-form expressions for the resistance and the inductance of different profiles of through-silicon vias, *IEEE Electron Device Lett.* 32 (2011) 393–395.
- [90] A. Nabil, J.A. Bernardo, Y. Mac, M. Abouelatta, A. Shaker, L.F. Bouchet, H. Ragai, C. Gontrand, Electrical modeling of tapered TSV including MOS-field effect and substrate parasitics: analysis and application, *Microelectron. J.* 100 (2020) 104797.
- [91] F. Wang, Z. Zhu, Y. Yang, X. Liu, R. Ding, Capacitance characterization of tapered through-silicon-via considering MOS effect, *Microelectron. J.* 45 (2014) 205–210.
- [92] Z.A.S. Mohammed, M.A.S. Olimpo, D.P. Poenar, S. Aditya, Smoothing of scalloped DRIE trench walls, *Mater. Sci. Semicond. Process.* 63 (2017) 83–89.
- [93] J.S. Park, D.H. Kang, S.M. Kwak, T.S. Kim, J.H. Park, T.G. Kim, S.H. Baek, B. C. Lee, Low-temperature smoothing method of scalloped DRIE trench by post-dry etching process based on SF₆ plasma, *Micro Nano Syst. Lett.* 8 (2020) 14.
- [94] C.M. Silvestre, V. Nguyen, H. Jansen, O. Hansen, Deep reactive ion etching of ‘grass-free’ widely-spaced periodic 2D arrays, using sacrificial structures, *Microelectron. Eng.* 223 (2020) 111228.
- [95] Q. Lu, Z. Zhu, Y. Yang, R. Ding, Y. Li, Electrical modeling and analysis of Cu-CNT heterogeneous coaxial through-silicon vias, *IEEE Trans. Nanotechnol.* 16 (2017) 695–702.
- [96] K. Rajkumar, G.U. Reddy, Performance comparison between copper and carbon nanotube based TSV for 3D-integrated circuits, *Mater. Today: Proc.* (2023), <https://doi.org/10.1016/j.matpr.2023.03.368> in press.
- [97] J. Su, R. Ma, X. Chen, L. Han, R. Yang, W. Zhang, Low-loss shielded through-silicon vias filled with multi-walled carbon nanotube bundle, *Microelectron. J.* 58 (2016) 83–88.
- [98] L. Yu, H. Yang, T.T. Jing, M. Xu, R. Geer, W. Wang, Electrical characterization of RF TSV for 3D multi-core and heterogeneous ICs, in: 2010 IEEE/ACM International Conference on Computer-Aided Design, 2010, pp. 686–693.
- [99] Z. Xu, J.Q. Lu, Three-dimensional coaxial through-silicon-via (TSV) design, *IEEE Electron Device Lett.* 33 (2012) 1441–1443.
- [100] K. Salah, Y. Ismail, Design of adiabatic TSV, SWCNT TSV, and air-gap coaxial TSV, in: 2015 IEEE International Symposium on Circuits and Systems, 2015, pp. 1953–1956.
- [101] B. Vaisband, A. Maurice, C.W. Tan, B.K. Tay, E.G. Friedman, Multi-bit CNT TSV for 3-D ICs, in: 2020 IEEE International Symposium on Circuits and Systems, 2020, pp. 1–5.
- [102] Y.F. Zhu, L. Shi, C. Zhang, X.Z. Yang, J. Liang, Preparation and properties of alumina composites modified by electric field-induced alignment of carbon nanotubes, *Appl. Phys. A Mater. Sci. Process.* 89 (2007) 761–767.
- [103] M. Matsumoto, R. Yamaguchi, K. Shima, M. Mukaida, M. Tomita, T. Watanabe, T. Ishida, T. Fujigaya, Control of anisotropic conduction of carbon nanotube sheets and their use as planar-type thermoelectric conversion materials, *Sci. Technol. Adv. Mater.* 22 (2021) 272–279.
- [104] W.S. Zhao, Q.H. Hu, Carbon nanotube through-silicon via: Modeling, design and applications, in: 2020 International Conference on Microwave and Millimeter Wave Technology, 2020, pp. 1–3.
- [105] Q.H. Hu, W.S. Zhao, D.W. Wang, G. Wang, Circuit modeling of shielded differential carbon nanotube bundle filled through-silicon vias, in: 2020 IEEE MTT-S International Conference on Numerical Electromagnetic and Multiphysics Modeling and Optimization, 2020, pp. 1–3.
- [106] Q.H. Hu, W.S. Zhao, K. Fu, D.W. Wang, G. Wang, On the applicability of two-bit carbon nanotube through-silicon via for power distribution networks in 3-D integrated circuits, *IET Circuits Devices Syst.* 15 (2021) 20–26.
- [107] A. Kumar, R. Chandel, R. Dhiman, Proposal and analysis of carbon nanotube based differential multibit through glass vias, *Microelectron. J.* 126 (2022) 105500.

- [108] C. Subramaniam, T. Yamada, K. Kobashi, A. Sekiguchi, D.N. Futaba, M. Yumura, K. Hata, One hundred fold increase in current carrying capacity in a carbon nanotube–copper composite, *Nat. Commun.* 4 (2013) 2202.
- [109] L. Ladani, J. Razmi, M. Sadeghilaridjani, Fabrication of cu-CNT composite and cu using laser powder bed fusion additive manufacturing, *Powders I* (2022) 207–220.
- [110] S. Prabhu, R. Ambigai, B.K. Vinayagam, Thermal and surface analysis of copper–CNT and copper–graphene-based composite using Taguchi–Grey relational analysis, *Aust. J. Mech. Eng.* 19 (2021) 95–106.
- [111] K.D. Bharathi, M.R. Rahman, A.K. Yadav, B.V.R. Murthy, P.D. Bhat, Optimized heat transfer rate in Cu/CNT nano composite prepared by electrodeposition technique, *ECS Adv.* 2 (2023) 011001.
- [112] D.J. McIntyre, R.K. Hirschman, I. Puchades, B.J. Landi, Enhanced copper–carbon nanotube hybrid conductors with titanium adhesion layer, *J. Mater. Sci.* 55 (2020) 6610–6622.
- [113] M.B. Jordan, Y. Feng, S.L. Burkett, Development of seed layer for electrodeposition of copper on carbon nanotube bundles, *J. Vac. Sci. Technol. B* 33 (2015) 021202.
- [114] Y. Chai, K. Zhang, M. Zhang, P.C.H. Chan, M.M.F. Yuen, Carbon nanotube/copper composites for via filling and thermal management, in: 2007 Proceedings 57th Electronic Components and Technology Conference, 2007, pp. 1224–1229.
- [115] J. Lee, S. Berrada, F. Adamu-Lema, N. Nagy, V.P. Georgiev, T. Sadi, J. Liang, R. Ramos, H. Carrillo-Nunez, D. Kalita, K. Lilienthal, M. Wislicenus, R. Pandey, B. Chen, K.B.K. Teo, G. Goncalves, H. Okuno, B. Uhlig, A. Todri-Sanial, J. Dijon, A. Asenov, Understanding electromigration in Cu-CNT composite interconnects: a multiscale electrothermal simulation study, *IEEE Trans. Electron Dev.* 65 (2018) 3884–3892.
- [116] Y. Feng, S.L. Burkett, Modeling a copper/carbon nanotube composite for applications in electronic packaging, *Comput. Mater. Sci.* 97 (2015) 1–5.
- [117] K. Fu, J. Zheng, W.S. Zhao, Y. Hu, G. Wang, Analysis of transmission characteristics of copper/carbon nanotube composite through-silicon via interconnects, *Chin. J. Electron.* 28 (2019) 920–924.
- [118] Y. Feng, S.L. Burkett, Fabrication and electrical performance of through silicon via interconnects filled with a copper/carbon nanotube composite, *J. Vac. Sci. Technol. B* 33 (2015) 022004.
- [119] S. Sun, W. Mu, M. Edwards, D. Mencarelli, L. Pierantoni, Y. Fu, K. Jeppson, J. Liu, Vertically aligned CNT-Cu nano-composite material for stacked through-silicon-via interconnects, *Nanotechnology* 27 (2016) 335705.
- [120] G. Chen, R. Sundaram, A. Sekiguchi, K. Hata, D.N. Futaba, Through-silicon-via interposers with Cu-level electrical conductivity and Si-level thermal expansion based on carbon nanotube-Cu composites for microelectronic packaging applications, *ACS Appl. Nano Mater.* 4 (2021) 869–876.
- [121] M. Rao, Electrical modeling and characterization of copper/carbon nanotubes in tapered through silicon vias, in: 2017 30th International Conference on VLSI Design and 2017 16th International Conference on Embedded Systems, 2017, pp. 366–371.
- [122] M. Rao, Electrical modeling of copper and mixed carbon bundles as a composite for 3D interconnect applications, in: 2020 IEEE 15th International Conference on Nano/Micro Engineered and Molecular System, 2020, pp. 495–499.
- [123] Y. Nadir, H. Belahrach, A. Ghammaz, A. Naamane, M. Radouani, Modeling crosstalk effects of hybrid copper carbon nanotube interconnects using a novel accurate FDTD based method, *Microelectron. J.* 129 (2022) 105589.